Linear Operators and Spectral Theory

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This elaboration is a review on the theory of linear operators acting in complex Banach or Hilbert spaces and their spectral properties. The subject belongs to the mathematical area of *functional analysis* and has many useful applications, especially to linear partial differential operators of elliptic and parabolic type and their associated boundary and initial value problems.

By use of spectral theory one can study resonance phenomena appearing in mathematical physics and engineering science. Here we are led to so-called *eigenvalue problems*, that is, equations of the form $\mathcal{A}\,u = \lambda\,u$ with $\mathcal{A}$ being a linear-elliptic partial differential operator, where the question arises, for which complex parameters $\lambda \in \mathbb{C}$ this equation has non-trivial solutions. But spectral theory also forms the foundation for *semigroup theory*, which is a decicional tool to solve evolution equations of the form $\dot{u} + \mathcal{A}\,u = f$ and $\ddot{u} + \mathcal{A}\,u = f$, respectively.

I wrote this treatise in honour of Prof. Dr. P. Werner (1932-2018), who conveyed my fascination for these beautiful mathematical topics by his lessons hold in 1984 and 1985 at Mathematical Institute A of University Stuttgart.

1. Banach and Hilbert Spaces

Throughout this text let $\mathbb{K}$ be one of the fields $\mathbb{R}$ and $\mathbb{C}$. By a *normed space* X we mean a $\mathbb{K}$-linear space that differs from other abstract linear spaces by the presence of a particular mapping $\|\cdot\| : X \to \mathbb{R}$, denoted the *norm* on X, which fulfills for all $x, y \in X$ and $\lambda \in \mathbb{K}$

(N-1) $\|x\| \geq 0$,

(N-2) $\|\lambda\, x\| = |\lambda| \cdot \|x\|$,

(N-3) $\|x + y\| \leq \|x\| + \|y\|$ *(triangle inequality)*,

(N-4) $\|x\| = 0$ if and only if $x = \mathsf{0}$.

Linear spaces may of course be furnished with different norms. We shall call arbitrary norms $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ on a linear space X to be *equivalent* to each another, if there are constants $c_1, c_2 > 0$ such that $c_1\,\|x\|_\alpha \leq \|x\|_\beta \leq c_2\,\|x\|_\alpha$ for all $x \in X$.

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Each normed linear space X can be furnished with a very natural topology $\mathcal{T}$, called the *norm topology* on X. Its definition is given either by the assertion that $\mathcal{T} := \mathcal{T}_{\|\cdot\|}$ is the smallest topology on X such that the norm mapping remains continuous, or by the metric topology $\mathcal{T}_d$, when X is considered as a metric space with metric function $d(x, y) := \|x - y\|$ for all $x, y \in X$. In fact, both definitions lead to the same collection of so-called "open" subsets in X, so we have $\mathcal{T} := \mathcal{T}_{\|\cdot\|} = \mathcal{T}_d$, which justifies to speak of *the* topology of a normed linear space. Hence a multiplicity of topological concepts, like the *closure* of a set, the *neighbourhood* of a point, the *convergence* of a series and many others are meaningful in the setting of normed linear spaces. Notice that equivalent norms on X are creating one and the same topology. For example, if X is a finite-dimensional linear space, then all norms are equivalent and consequently there is exactly one topology on X induced by them.

Recall from metric space theory that a sequence (x_n) in a normed linear space X is called *Cauchy*, if to each $\epsilon > 0$ there is $N_\epsilon \in \mathbb{N}$ such that $m, n > N_\epsilon$ implies $\|x_m - x_n\| < \epsilon$. The normed space X is called *sequentially complete* (or shortly *complete*), if for any Cauchy sequence (x_n) there is $x \in X$ such that $\lim_{n\to\infty} \|x_n - x\| = 0$. Equivalently, X is complete iff for every absolutely convergent series $\sum_{n=1}^{\infty} \|x_n\| < \infty$ there is $x \in X$ such that $\|x - x_n\| \to 0$ for $n \to \infty$. A complete normed linear space is called a *Banach space*.

It should be mentioned that each uncomplete normed linear space can be regarded as a dense linear subspace of a certain Banach space. The general procedure readily follows the way how to construct the set of real numbers out of that of the rational ones.

We go ahead with the notion of *embedding*, which is a special kind of mappings between normed linear spaces and leads to the concept of norm-isomorphism. Preliminary, a mapping $f : X \to Y$ is called continuous if for every open set $\mathcal{O} \subset Y$ the pre-image $f^{-1}(\mathcal{O})$ is open in X. Then by an embedding ι is meant a continuous and injective mapping from a normed linear space X into a further normed linear space Y, which is *linear*, that is,

$$\iota(x + y) = \iota x + \iota y \quad \text{and} \quad \iota(\lambda x) = \lambda \iota x$$

holds for $x, y \in X$ and $\lambda \in \mathbb{K}$. If ι is even surjective, it is said to be a linear isomorphism between the spaces X and Y. Then BANACH*'s isomorphism theorem* claims that the inverse mapping ι^{-1} is also an embedding, so ι and ι^{-1} both constitute linear homeomorphisms between X and Y. Furthermore, the embedding ι is said to be a *norm-isomorphism*, if it is surjective and $\|\iota x\| = \|x\|$ holds for all $x \in X$ (notation: $X \cong Y$). It should be clear that in this case the inverse mapping ι^{-1} is a norm-isomorphism, too.

A complex (real) *Hilbert space* differs from an arbitrary Banach space by the presence of an *inner product*, which is nothing but a symmetric and positive sesquilinear (bilinear) form on such a space. This form defines the norm and the metric topology of the Hilbert space, but also determines the inner geometry to be close to that of an Euclidean space.

We want to make these assertions precise. Given a real or complex linear space H, an *inner product* in H is defined to be a mapping $\langle\,\cdot\,,\,\cdot\,\rangle : H \times H \to \mathbb{K}$, which is

(I-1) linear in the first argument,
(I-2) either linear or antilinear in the second argument (conditional upon, whether $\mathbb{K}$ is the real or the complex number field),

together with the additional properties

(I-3) $\langle y, x\rangle = \langle x, y\rangle$ ($\mathbb{K} = \mathbb{R}$) rsp. $\langle y, x\rangle = \overline{\langle x, y\rangle}$ ($\mathbb{K} = \mathbb{C}$),
(I-4) $\langle x, x\rangle \geq 0$ for all $x \neq 0$,
(I-5) $\langle x, x\rangle = 0$ implies $x = 0$.

Property (I-3) is denoted the *Hermitean symmetry* of $\langle\,\cdot\,,\,\cdot\,\rangle$, while (I-4) and (I-5) together mean *positive definiteness.* The presence of the inner product gives rise to define a norm in H according to

$$\|x\| := \langle x, x\rangle^{1/2} \qquad (x \in H), \tag{1.1}$$

so each linear space endowed with an inner product is normed in a natural manner. The inner product and the norm are coupled by the *Cauchy-Schwarz inequality*

$$|\langle x, y\rangle| \leq \|x\| \cdot \|y\| \qquad (x, y \in H),$$

which implies that the mapping $\langle\,\cdot\,,\,\cdot\,\rangle : H \times H \to \mathbb{K}$ is continuous. An inner product space H is called a *Hilbert space* or shortly *Hilbertian*, if it is complete with respect to the norm defined by (1.1).

The pleasant geometric properties of inner product spaces may be attributed to the validness of the *parallelogram equation*

$$2\,\|x\|^2 + 2\,\|y\|^2 = \|x + y\|^2 + \|x - y\|^2 \qquad (x, y \in H). \tag{1.2}$$

In general, the norm $\|\cdot\|$ on a linear space X can be induced by an inner product if and only if (1.2) holds. In this case and if X is a complex linear space, the inner product can be recovered from the norm by the *polarization formula*

$$\langle x, y\rangle = \frac{1}{4}\left(\|x+y\|^2 - \|x-y\|^2 + i\,\|x+iy\|^2 - i\,\|x-iy\|^2 \right).$$

If X is a real space, however, the polarization formula simplifies to

$$\langle x, y\rangle = \frac{1}{4}\left(\|x+y\|^2 - \|x-y\|^2 \right).$$

In the following we want to point out that Hilbert spaces differ from general Banach spaces by a couple of desirable properties:

1. *Orthogonality.* Two elements $x, y \in H$ are called *orthogonal* to each another, if $\langle x, y\rangle = 0$ (notation: $x \perp y$). Any subset $\mathfrak{O} \subset H$ is said to be *orthogonal* or an *orthogonal system*, if all elements in $\mathfrak{O}$ are orthogonal to each other. Finally, the *orthogonal space* $\mathcal{A}^{\perp}$ of a subset $\mathcal{A} \subset H$ contains exactly the elements $y \in H$, which are orthogonal to all $x \in \mathcal{A}$, and forms a closed linear subspace of H.

2. *Projection Theorem.* Let $\mathcal{A} \subset H$ be a *closed* linear subspace. Then for each $x \in H$ there is exactly one $x_0 \in \mathcal{A}$ such that

$$\|x - x_0\| = \inf\{\, \|x - y\| : y \in \mathcal{A} \,\}.$$

The element x_0 is called the *projection* of x onto $\mathcal{A}$ and minimizes the distance between $\mathcal{A}$ and x. The associated linear mapping $\pi_{\mathcal{A}} : x \in H \mapsto x_0 \in \mathcal{A}$ is also denoted the *projection* of H onto $\mathcal{A}$.

3. *Decomposition Theorem.* Using the notation of orthogonal space $\mathcal{A}^{\perp}$, the projection of an element $x \in H$ onto $\mathcal{A}$ is characterized as follows: $x_0 = \pi_{\mathcal{A}} x$ iff $x - x_0 \in \mathcal{A}^{\perp}$. Hence each $x \in H$ can uniquely be represented by the formula $x = x_0 + x^{\perp}$, where $x_0 \in \mathcal{A}$ and $x^{\perp} \in \mathcal{A}^{\perp}$, so the entire space H is norm-isomorphic to the direct sum $\mathcal{A} \oplus \mathcal{A}^{\perp}$ of the closed linear subspaces $\mathcal{A}$ and $\mathcal{A}^{\perp}$. This fact is known as the *decomposition theorem* in Hilbert space.

A closed linear subspace $\mathcal{A}$ of a Banach space X is said to be *topologically complementable*, if there is a closed subspace $\mathcal{B} \subset X$ such that $\mathcal{A} \cap \mathcal{B} = \{0\}$, $[\mathcal{A} \cup \mathcal{B}] = X$ and $X \cong \mathcal{A} \oplus \mathcal{B}$, that is, the space X is linear-homeomorphic to the product space $\mathcal{A} \times \mathcal{B}$. Using this notation, the decomposition theorem quotes that in the Hilbert space setting every closed subspace $\mathcal{A}$ is topologically complemented by the orthogonal space $\mathcal{A}^{\perp}$.

4. *Existence of Orthonormal Bases.* An orthogonal system $\mathfrak{O} \subset H$ is denoted an *orthonormal system*, if $\langle x, x\rangle = 1$ for all $x \in \mathfrak{O}$. However, if $\mathfrak{O}$ contains only *finitely* many elements, then the span $[\mathfrak{O}]$ forms a *closed* subspace in H, and the projection $\pi_{[\mathfrak{O}]}$ of any element x onto $[\mathfrak{O}]$ can be expressed by the formula

$$\pi_{[\mathfrak{O}]}\, x \;=\; \sum_{i=1}^{n} \langle x, e_i\rangle\, e_i .$$

But if $\mathfrak{O}$ consists of arbitrarily many elements, then the inequality $\langle x, e_\alpha\rangle \neq 0$ can only be valid for mostly countable many elements $e_\alpha \in \mathfrak{O}$, which in conclusion enables to add the coefficients $\langle x, e_\alpha\rangle$ together about all elements of $\mathfrak{O}$.

An orthonormal system $\mathfrak{O}$ is called an *orthonormal base*, if its span $[\mathfrak{O}]$ lies dense in H. In this case it is possible to represent each element $x \in H$ by the formal sum

$$x \;=\; \sum_{e_\alpha \in \mathfrak{O}} \langle x, e_\alpha\rangle\, e_\alpha . \tag{1.3}$$

The inner products $\langle x, e_\alpha\rangle$ are called the *Fourier coefficients* of x with respect to $\mathfrak{O}$. Notice that the right hand side of (1.3) consists of *mostly countable many* non-zero summands, and this series proves to be invariant by realignments of the elements $e_\alpha \in \mathfrak{O}$.

There is of course the need to determine, whether a given orthonormal system $\mathfrak{O}$ forms an orthonormal base. In fact, it can be shown that the following statements are equivalent:

(a) $\mathfrak{O}$ is an orthonormal base in H.
(b) $\mathfrak{O}$ is *complete*, that is, $\langle x, e\rangle = 0$ for all $e \in \mathfrak{O}$ implies $x = 0$.
(c) $\sum_{e_\alpha \in \mathfrak{O}} |\langle x, e_\alpha\rangle|^2 = \|x\|^2$ for all $x \in H$ (*Parseval's equation*).

It is also nearby to ask for the existence of orthonormal bases. First of all, the collection of all orthonormal systems in a Hilbert space H forms a partial ordered set, and under the assumption that ZORN's lemma[1] is valid, it contains a *maximal* element $\mathfrak{O}$ that cannot be increased, hence each Hilbert space $H \neq \{0\}$ owns an orthonormal base.

Furthermore, if $\mathfrak{O}_1$ and $\mathfrak{O}_2$ are orthonormal bases in H, then there is a bijective mapping between them, thus they own identical

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1 The *axion of choice* claims that if $\mathcal{F}$ is a disjoint family F_α $(\alpha \in I)$ of non-empty sets, then there is a set A that shares exactly one element with each other set $F_\alpha \in \mathcal{F}$.

cardinal numbers. This legitimates us to introduce the notation of *Hilbert space dimension* $|H|$ being the cardinal number of any orthonormal base.

A Hilbert space H is called *separable* if there is a finite or mostly countable subset lying dense in H. It turns out that H is separable if and only if $|H| \leq |\mathbb{N}|$. Indeed, in this case every orthonormal base in H is finite or mostly countable, and the space H itself proves to be norm-isomorphic to the Hilbert space ℓ^2 of square-summarizable sequences $f : \mathbb{N} \to \mathbb{K}$.

2. Basic Theory of Linear Operators

In this section we want to sketch the fundamentals of operator theory. If we assume X, Y to be normed linear spaces, then by a *linear operator* between X and Y is meant a mapping $A : \mathfrak{D}(A) \subset X \to Y$, where the *domain of definition* $\mathfrak{D}(A)$ forms a linear subspace of X and A is *linear*, that is,

$$A(\lambda x + \mu y) \;=\; \lambda\, Ax + \mu\, Ay$$

holds for all $x, y \in \mathfrak{D}(A)$ and $\lambda, \mu \in \mathbb{K}$. The space X is denoted the *domain space* of A, whereas Y is said to be the *range space* of A.

The domain of definition $\mathfrak{D}(A)$ may of course coincide with the entire space X, but it is often sufficient to require that $\mathfrak{D}(A)$ lies dense in X, so each $x \in X$ can be approximated by a sequence of elements in $\mathfrak{D}(A)$. In this case it is usual to say that A is *densely defined* in X. However, if the set $\mathfrak{D}(A)$ is *not* dense in X, it is nearby to consider the closure X_0 of $\mathfrak{D}(A)$ in X and to pay attention to the slightly modified but densely defined operator $A_0 : \mathfrak{D}(A) \subset X_0 \to Y$.

Rather the domains of definition for operators A_1 and A_2 with same domain space X and range space Y may be proper linear subspaces of X, the *sum* $A_1 + A_2$ and the *composition* $A_1 \circ A_2$ are well-defined, if their domains of definition are given by

$$\begin{cases} \mathfrak{D}(A_1 + A_2) \;:=\; \mathfrak{D}(A_1) \cap \mathfrak{D}(A_2), \\ \mathfrak{D}(A_1 \circ A_2) \;:=\; \{\, x \in \mathfrak{D}(A_2) : A_2\, x \in \mathfrak{D}(A_1) \,\}. \end{cases}$$

Furthermore, for arbitrary $\lambda \in \mathbb{K}$ the operator λA is well-defined by $\mathfrak{D}(\lambda A) := \mathfrak{D}(A)$ and $(\lambda A)x := \lambda(Ax)$ for $x \in \mathfrak{D}(\lambda A)$ and is called a *dilatation* in X.

If the domain of definition is a proper subset of the domain space, the concept of *extension* is meaningful. Let A_1 and A_2 be linear operators with domain of definitions $\mathfrak{D}(A_1) \subset X$ and $\mathfrak{D}(A_2) \subset X$, respectively. We introduce the partial ordered relation

$$A_1 \prec A_2 :\Longleftrightarrow \mathfrak{D}(A_1) \subset \mathfrak{D}(A_2) \ \text{ and } \ A_1\,x = A_2\,x \text{ for all } x \in \mathfrak{D}(A_1)$$

and denote A_2 an *extension* of A_1 if $A_1 \prec A_2$. The operators A_1 and A_2 are said to be *equal* (briefly $A_1 = A_2$) iff $A_1 \prec A_2$ and $A_2 \prec A_1$, or equivalently, if $\mathfrak{D}(A_1) = \mathfrak{D}(A_2)$ and $A_1\,x = A_2\,x$ for all $x \in \mathfrak{D}(A_1)$.

Finally, to every linear operator $A : \mathfrak{D}(A) \subset X \to Y$ there are basically assigned three linear subspaces, namely

1. the *kernel* $\mathfrak{N}(A) := \{\, x \in \mathfrak{D}(A) : Ax = 0 \,\} \subset X$;
2. the *range* $\mathfrak{R}(A) := \{\, Ax : x \in \mathfrak{D}(A) \,\} \subset Y$;
3. the *cokernel* $\mathfrak{C}(A) \subset Y$, that is, the linear-algebraic complement of $\mathfrak{R}(A)$ within the space Y, hence $\mathfrak{R}(A) \oplus \mathfrak{C}(A) = Y$.

A first classification of linear operators between a domain space X and a range space Y can be performed by checking certain properties, which are defined by means of the norm topologies in these spaces. It leads in conclusion to the concepts of *continuous*, *compact*, *closed* and *closable* operator.

1. Remember that a linear operator $A : \mathfrak{D}(A) \subset X \to Y$ is *continuous*, if the pre-image $A^{-1}(\mathcal{O})$ of each open set $\mathcal{O} \subset Y$ is open in the domain space X. However, in the setting of normed linear spaces we may replace continuity by *sequential continuity*, which means that A is continuous in $x \in X$ if for each convergent series (x_n) with limit element $x \in X$ we have $\lim_{n\to\infty} Ax_n = Ax$.

   Moreover, a linear operator A between normed linear spaces X and Y is denoted *norm-bounded* or simply *bounded*, if there is $c \geq 0$ such that $\|Ax\| \leq c \cdot \|x\|$ for each $x \in \mathfrak{D}(A)$. It is easy to check that A is continuous iff A is bounded, hence these concepts are equivalent to each other. In this case the *operator norm* of A is well-defined according to

$$\|A\| := \sup_{x \in \mathfrak{D}(A), \|x\| = 1} \|Ax\| \,. \tag{2.1}$$

   If a linear operator A is continuous and $\mathfrak{D}(A)$ is a dense subset of the domain space X, then A can uniquely be extended to a linear mapping defined on the closure $\overline{\mathfrak{D}(A)} = X$, which proves to be continuous as well. This is the assertion of the *B.L.T. theorem* (written out the *bounded linear transformation theorem*, see [14], page 6). It is founded by the fact that any continuous linear operator A is even *uniformly* continuous, that is, for each $\epsilon > 0$ there exists $\delta > 0$, such that

$$\|x - y\|_X < \delta \text{ implies } \|Ax - Ay\|_Y < \epsilon$$

for $x, y \in \mathfrak{D}(A)$, hence δ can be chosen independently of x, y. Thus we may basically assume that if A is continuous, then $\mathfrak{D}(A) = X$ by applying this canonical extension, and the operator A finally appears as a mapping $A : X \to Y$.

Let $\mathscr{L}(X, Y)$ be the set of *bounded* linear operators $A : X \to Y$. Endowed with addition $A+B$ and scalar multiplication λA, the set $\mathscr{L}(X, Y)$ forms a linear space, and furnished with the norm given by (2.1) it is even a normed linear space. This space is complete, if Y is so, and in this case $\mathscr{L}(X, Y)$ forms a Banach space in its own right.

If $Y = X$, the set $\mathscr{L}(X) := \mathscr{L}(X, X)$ constitutes a *normed algebra* with composition $(A_1, A_2) \mapsto A_2 \circ A_1$ as its multiplicative operation and owns the property

$$\|A_1 \circ A_2\| \leq \|A_1\| \cdot \|A_2\| .$$

It is called the *operator algebra* associated with X and actually forms a Banach algebra if Y is complete.

Let us collect the obtained results to the following main theorem on continuous linear operators, where X and Y are assumed to be normed linear spaces:

THEOREM 2.1. For $A : X \to Y$ being a linear operator, the subsequent assertions are equivalent to each other:

(a) A is continuous;
(b) A is uniformly continuous;
(c) A is sequentially continuous in every $x \in X$;
(d) A is sequentially continuous in the origin $0 \in X$;
(e) A is bounded.

2. A linear operator $k : X \to Y$ between Banach spaces X, Y is called *compact*, if the image of each bounded set under k is relative compact, or equivalently, if $\{x_n\}_{n\in\mathbb{N}}$ is a bounded sequence in X, then the sequence $\{k\, x_n\}_{n\in\mathbb{N}}$ in Y owns a convergent subsequence.

For example, a linear operator k is compact, if at least one of the spaces X and Y is finite-dimensional. In case of $Y = X$, the identity operator $1_X : X \to X$ is compact iff X is finite-dimensional.

Compact operator own a couple of useful properties:

(a) every compact linear operator is continuous;
(b) the composition $A_1 \circ A_2$ of bounded linear operators is compact, if one of them is so;
(c) the first and second property together imply that the set $\mathscr{C}(X)$ of compact linear operators $k : X \to X$ is a *closed* ideal within the operator algebra $\mathscr{L}(X)$, hence it is complete and forms a Banach algebra in its own right;
(d) finite linear combinations of compact linear operators are compact, too;
(e) the limit of a norm convergent sequence of compact operators is compact.

A linear operator $f : X \to Y$ is said to have *finite rank*, if the range $\mathfrak{R}(f)$ is finite-dimensional. It is obvious that f is compact.

Suppose that (f_n) is a sequence of finite-rank operators converging to the operator A with respect to the norm of $\mathscr{L}(X, Y)$. Then A is compact and the set $\mathscr{C}(X, Y)$ of compact linear operators proves to be a subset of the closure of the set $F(X, Y)$ of finite-rank operators, i. e. $\mathscr{C}(X, Y) \subset \overline{F(X, Y)}$.

Reversely, in the setting of Hilbert spaces each compact linear operator k appears in this way, that is, k is the norm-limit of a sequence of finite-rank operators (but this is not necessarily true in the general setting of Banach spaces).

Let $\mathfrak{B} = \{e_1, e_2, \ldots\}$ be any orthonormal base of a separable Hilbert space H and let $A \in \mathscr{L}(H)$. We define the number $S(A)$ according to

$$S(A) := \sum_{i \in \mathfrak{B}} \|A\, e_i\|^2 .$$

Then it turns out this definition for $S(A)$ does not depend on the choice of the orthonormal base. In case of $S(A) < \infty$ the operator A is called a HILBERT-SCHMIDT *operator*, and the number

$$\|A\|_{HS} := \sqrt{S(A)}$$

is denoted the *Hilbert-Schmidt norm* of A, since the mapping $A \mapsto \|A\|_{HS}$ owns the properties of a norm.

The set $S(H)$ of Hilbert-Schmidt operators in H is topologically and algebraically closed in $\mathscr{L}(H)$ with respect to the operations $A_1 + A_2$ and λA, hence $S(H)$ endowed with norm $\|\cdot\|_{HS}$ forms a

Banach space in its own right. Each Hilbert-Schmidt operator is compact, so $S(H)$ forms a two-sided ideal within the space $\mathscr{C}(H)$ of compact linear operators in H.

3. A linear operator $A:\mathfrak{D}(A)\subset X\to Y$ is called *graph-closed* (or shortly *closed*) if the set

$$\Gamma_A := \{(x,Ax): x\in\mathfrak{D}(A)\} \subset X\times Y,$$

denoted the *graph* of A, is closed in the space $X\times Y$ with respect to the product topology $\mathcal{T}_{X\times Y}$. This property especially implies that the kernel $\mathfrak{N}(A)$ of A forms a closed set in X.

It is a matter of fact that an operator A is closed if and only if it is *sequentially closed*, that is, $x_n\to x$ and $Ax_n\to y$ implies $x\in\mathfrak{D}(A)$ and $Ax=y$, briefly written

$$\left.\begin{array}{r} x_n\to x \\ Ax_n\to y \end{array}\right\} \quad\Longrightarrow\quad x\in\mathfrak{D}(A) \text{ and } Ax=y.$$

This can easily proved by the fact that a subset of a normed linear space is closed if and only if it is sequentially closed. Checking the sequentially closeness is the usual method to verify that a given linear operator is graph-closed. But notice that the equivalence of these both notations is in general not valid, if X and Y are non-metrizable topological linear spaces.

It is often helpful to make the domain of definition $\mathfrak{D}(A)$ of a linear operator A into a normed linear space by its own. This can be done by introducing the so-called *graph norm*

$$\|x\|_A := \|x\|_X + \|Ax\|_Y$$

for $x\in\mathfrak{D}(A)$. In fact, the pair $(\mathfrak{D}(A),\|\cdot\|_A)$ forms a normed linear space, which constitutes a Banach space if and only if A is sequentially closed. This equivalence provides a further method to verify closeness of a given linear operator.

The following theorem uses the possible validness of an abstract *Schauder estimate* to ensure sequentially closenss:

THEOREM 2.2. Let $A_0: X\to Y$ be a continuous linear operator, where the Banach space X is continuously and densely embedded into the Banach space Y by a mapping $\iota: X\to Y$. If the estimate

$$\|x\|_X \le c\cdot\{\,\|\iota x\|_Y + \|A_0\,x\|_Y\,\} \tag{2.2}$$

is fulfilled for some $c>0$ and all $x \in X$, then the linear operator A acting in Y with $\mathfrak{D}(A) := \iota(X)$ and $A\,\iota x := A_0\,x$ for $x \in X$ is closed.

In fact, let $(x_n)_{n\in\mathbb{N}}$ be a sequence in $\mathfrak{D}(A)$ with $\|x_n - x\|_X \to 0$ and $\|A\,x_n - y\|_Y \to 0$. It is to verify that $x \in \mathfrak{D}(A)$ and $Ax = y$. Inequality (2.2) implies that $(x_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in X with limit element $x \in X$. Since A_0 is continuous, the element y is also the limit of the sequence $(A\,\iota x_n)_{n\in\mathbb{N}}$, hence A is closed.

4. Many linear operators acting between Banach or Hilbert spaces are not closed, but can be extended to operators being closed. We shall call such operators *closable* or *pre-closed.* If a linear operator A is closable, then there is a *minimal closed extension* A_c of A, called the *closure* of A. This mapping A_c is given by the fact that each other closed extension of A extends A_c, and A_c is nothing but the intersection of all closed extensions of A. Clearly, each closed operator is closable, since it constitutes a closed extension of itself. Furthermore, given a closed linear operator A with domain of definition $\mathfrak{D}(A)$, any linear subspace $\mathcal{L} \subset \mathfrak{D}(A)$ is called a *core* of A, if the closure of A restricted to $\mathcal{L}$ resembles A.

   There is a well-known criterion for closability:

   THEOREM 2.3. A linear operator A with domain of definition $\mathfrak{D}(A)$ is closable, if each sequence $(x_n, Ax_n)_{n\in\mathbb{N}}$ with $x_n \in \mathfrak{D}(A)$ for $n \in \mathbb{N}$ and converging to $(0, y)$ implies $y = 0$, briefly written

$$\left.\begin{array}{r} x_n \to 0 \\ Ax_n \to y \end{array}\right\} \quad \Longrightarrow \quad y = 0.$$

Let us emphasize that in general the concepts of closeness and continuity of a linear operator A are not comparable. However, if X and Y are Banach or Hilbert spaces and $\mathfrak{D}(A)$ is a *closed* set in X, then closeness of A is an easy consequence of continuity of A. This is especially true in the following situation: if A is densely defined and continuous, then A can be extended to a continuous linear operator $A_{ext} : X \to Y$, and since X itself is closed, the extension A_{ext} is a closed operator. Hence continuous linear operators of the form $A : X \to Y$ fit in a natural way into the class of densely defined and closed linear operators.

Reversely, the *closed graph theorem* claims that if $\mathfrak{D}(A)$ is closed, then closeness implies continuity, so both concepts are equivalent to each other under this restrictive condition.

The following application of the closed graph theorem is of outstanding meaning: if a linear operator A is closed and bijective, then its inverse A^{-1} is also linear and closed, and since the domain of definition $\mathfrak{D}(A^{-1})$ is the entire space Y, the closed graph theorem implies that $A^{-1}: Y \to X$ is continuous and bounded, respectively. This fact is known as the *bounded inverse theorem* for closed linear operators. But notice that the quotation is also valid for continuous linear operators $A: X \to Y$, which is an immediate consequence of BANACH's *open mapping theorem.*

In conclusion, if the domain of definition $\mathfrak{D}(A)$ of a linear operator A is a *closed* subspace of the domain space X, then the following chain of implications is valid:

$$A \text{ compact} \implies A \text{ bounded} \iff A \text{ continuous} \iff A \text{ closed.}$$

3. Adjoint Spaces and Operators

So far we have described and classified linear operators by various topological properties. Another approach to setup a classification in the totality of linear operators between a domain space X and a range space Y is to consider specific choices of the range space. For example, one can chose

1. $Y = X$ to obtain operators of the form $A: \mathfrak{D}(A) \subset X \to X$, and in this case the operator A is said to *act in* X. In fact, if the domain space and the range space are the same, one can study perturbations of A of the form $A+\lambda 1_X$ (shortly denoted $A+\lambda$), where $\lambda \in \mathbb{K}$ and 1_X denotes the *identity operator* in X, and this is performed in what is called *spectral theory* (see section 4).
2. $\mathfrak{D}(A) = X$ and $Y = \mathbb{K}$, leading to *linear functionals* $A: X \to \mathbb{K}$. By the *adjoint space* X^* of a normed linear space X is meant the linear space of continuous linear functionals $\phi: X \to \mathbb{K}$, which itself forms a Banach space with norm
$$\|\phi\| := \sup_{\|x\|=1} |f(x)| \,.$$
The *Hahn-Banach theorem* claims that $X^* \neq \{0\}$, so there are non-trivial continuous linear functionals on X. Any Banach space

X is called *reflexive*, if X is norm-isomorphic to the *bi-adjoint space* $X^{**} := (X^*)^*$.

3. $\mathfrak{D}(A) = X$ and $Y = X^*$, the *adjoint space* of X, so the linear operator adopts the form $A : X \to X^*$.

   In case of $X = H$, where H is a complex Hilbert space, it is more convenient to consider linear operators of the form $A : H \to H_a^*$, where H_a^* is the linear space of continuous and *conjugate linear* functionals $\phi : H \to \mathbb{C}$, which distinguish by the property

$$\phi(\lambda \cdot x) = \overline{\lambda}\, \phi(x) \quad \text{for all } x \in H \text{ and } \lambda \in \mathbb{C}.$$

   The linear space H_a^* is denoted the *conjugate space* of H and can be endowed with the norm

$$\|\phi\| := \sup_{\|x\|=1} |\phi(x)|\,.$$

   For each operator A with domain space H and range space H_a^* we can associate a *sesquilinear form* $\mathfrak{a}_A : H \times H \to \mathbb{C}$ by

$$\mathfrak{a}(x,y) := (Ax)y \qquad (x, y \in H). \tag{3.1}$$

   Reversely, if a sesquilinear form $\mathfrak{a}$ is given, we may introduce a linear operator $A_{\mathfrak{a}} : H \to H_a^*$ by the same formula (3.1), called the *representation operator* of $\mathfrak{a}$.

   Similar to theorem 2.1, the following theorem characterizes continuity of sesquilinear forms by equivalent assertions:

   THEOREM 3.1. Let $\mathfrak{a} : H \times H \to \mathbb{C}$ be a sesquilinear form. Then:
   (a) $\mathfrak{a}$ is continuous;
   (b) $\mathfrak{a}$ is sequentially continuous in (x, y) for every $x, y \in H$;
   (c) $\mathfrak{a}$ is sequentially continuous in $(0,0) \in H \times H$;
   (d) there is $\Theta \geq 0$ such that $|\mathfrak{a}(x,y)| \leq \Theta\, \|x\| \cdot \|y\|$ for $x, y \in H$;
   (e) The representation operator $A_{\mathfrak{a}}$ is continuous.

   Let us especially consider the case $\mathfrak{a}(x,y) = \langle x, y\rangle$. In this case the representation operator Φ of the inner product is called the *Riesz map* of H. Since inner products are positive definite by definition, the mapping Φ proves to be injective. But RIESZ*'s representation theorem* claims that Φ is also surjective, so there is a canonical correspondence between the elements of H and H_a^*:

THEOREM 3.2. If H is a complex Hilbert space, then $\Phi : H \to H_a^*$ is bijective and consequently a linear homeomorphism.

The proof of this theorem can be found in nearly every textbook on functional analysis.

Theorem 3.2 implies that for $\phi \in H_a^*$ there is a unique element $x_\phi \in H$, which fulfills the equation $\langle x_\phi, y\rangle = \phi(y)$ for every $y \in H$. Moreover, the Riesz map gives rise to introduce an inner product on the conjugate space H_a^* by

$$\langle \phi, \psi\rangle := \langle \Phi^{-1}(\phi), \Phi^{-1}(\psi)\rangle$$

for all $\phi, \psi \in H_a^*$, hence Φ is not only norm-preserving, but also *isometric*, that is, $\langle \Phi x, \Phi y\rangle = \langle x, y\rangle$ for $x, y \in H$.

On the other hand, if H is a *real* Hilbert space, we may replace H_a^* by H^* and verify that the Riesz map Φ again is a *norm-isomorphism* between the spaces H and H^*, since in this case the inner products in H and H^* are also linear in the second argument.

The representation theorem shows that real and complex Hilbert spaces are reflexive.

In operator theory it is often advantageous to combine the study of a densely defined linear operator with that of its *adjoint.* The rest of this section is a brief review on important concepts with respect to adjoint operators.

Let X, Y be arbitrary Banach spaces and X^*, Y^* be the adjoint spaces. If $A : \mathfrak{D}(A) \subset X \to Y$ is a linear operator with domain of definition $\mathfrak{D}(A)$ dense in X, then the domain of definition $\mathfrak{D}(A^*)$ of the adjoint operator A^* is defined by

$$\mathfrak{D}(A^*) := \{\, \phi \in Y^* : \phi \circ A : \mathfrak{D}(A) \to \mathbb{K} \text{ is continuous} \,\}.$$

For $\phi \in \mathfrak{D}(A^*)$ let $A^*\phi := \overline{\phi \circ A}$, the unique extension of the continuous linear functional $\phi \circ A$ to the whole of X, which implies $A^*\phi \in X^*$. Thus the linear operator A^* has domain space Y^* and range space X^*, and the domain of definition $\mathfrak{D}(A^*)$ forms a linear subspace of Y^*, on which A^* is linear.

We emphasize that A is supposed to be densely defined in order to introduce its adjoint. But the adjoint A^* is not necessarily densely defined. Indeed, it can even happen that $\mathfrak{D}(A^*) = \{0\}$. The question however, whether $\mathfrak{D}(A^*)$ is dense in X^*, depends essentially on the closability of the operator A, see assertion (d) in theorem 3.3 below.

The usage of the adjoint operator is founded by the validness of two relationships between the kernels and ranges of the operators A and A^*, namely

$$\overline{\mathfrak{R}(A^*)} = \mathfrak{N}(A)^\perp \quad \text{and} \quad \overline{\mathfrak{R}(A)} = {}^\perp\mathfrak{N}(A^*), \tag{3.2}$$

where for any subset $M \subset X$ and $N \subset X^*$ the *annihilator* $M^\perp$ and the *pre-annihilator* ${}^\perp N$ are defined by

$$\begin{cases} M^\perp := \{ x^* \in X^* : \langle x, x^* \rangle = 0 \text{ for all } x \in M \}, \\ {}^\perp N := \{ x \in X : \langle x, x^* \rangle = 0 \text{ for all } x^* \in N \} \end{cases}$$

and $\langle x, x^* \rangle := x^*(x)$ denotes the pairing of elements in X and X^*. Checking the validness of the relations (3.2) is elementary, hence we do not want to elaborate this in detail.

Adjoint operators probably enjoy a couple of properties:

THEOREM 3.3. Let X, Y be Banach spaces and A, B be densely defined linear operators with domain space X and range space Y.

(a) $A \prec B$ implies $B^* \prec A^*$.
(b) The adjoint operator A^* is closed.
(c) If A is closable, then $A^* = \overline{A}^*$.
(d) If A is closable, then A^* is densely defined, which implies that the bi-adjoint $A^{**} := (A^*)^*$ exists.

We also introduce the concept of *adjoint pair* of linear operators. The both densely defined linear operators $A : \mathfrak{D}(A) \subset X \to Y$ and $B : \mathfrak{D}(B) \subset Y^* \to X^*$ are said to be *formally adjoint* to each other or to be a *formally adjoint pair* (A, B) if

$$\langle A\,x, f \rangle_{Y \times Y^*} = \langle x, Bf \rangle_{X \times X^*}$$

holds for all $x \in \mathfrak{D}(A)$ and $f \in \mathfrak{D}(B)$.

In addition, we define the *restricted adjoint* B_r of the operator B according to $\mathfrak{D}(B_r) :=$

$$\{x \in X : \text{ there is } y \in Y \text{ such that } (Bf)\,x = f(y) \text{ for all } f \in \mathfrak{D}(B)\}$$

and $B_r\,x := y$. The mapping B_r acts between the Banach spaces X and Y and proves to be linear. We may reformulate the notation of formally adjoint pair by saying that A and B are formally adjoint to one another if B_r is an extension of A.

To the end, by a *realization* of a formally adjoint pair (A, B) we mean any densely defined and closed linear operator R lying between A and B_r, so $A \prec R \prec B_r$ must hold.

Recall that the adjoint A^* of a linear operator A forms a mapping from the adjoint space Y^* to the adjoint space X^*. With respect to the question of compactness of A and A^*, SCHAUDER*'s theorem* holds, whose proof can be found for example in [17], pp. 282-283:

THEOREM 3.4. A continuous linear operator $A: X \to Y$ between Banach spaces X and Y is compact if and only if A^* is compact.

In the Hilbert space setting, where $X = Y = H$ is assumed, the sets $M^{\perp}$ and ${}^{\perp}N$ are nothing but the orthogonal spaces of M and N, where $N^{\perp} \subset H^*$ can be seen as a linear subspace of H, which is a consequence of RIESZ's representation theorem (establishing a one-to-one relationship between the spaces H and H^*).

Furthermore, the adjoint A^* of a densely defined linear operator A can be regarded as a mapping acting in H. In fact, if A is linear and densely defined in H, then the domain of definition $\mathfrak{D}(A^*)$ of the adjoint A^* consists of those elements $y \in H$, for which there is $y^* \in H$ fulfilling

$$\langle A\,x, y\rangle = \langle x, y^*\rangle \text{ for all } x \in \mathfrak{D}(A).$$

We then define $A^* y := y^*$ and denote A^* the *Hilbert adjoint* of A.

Recall that if a linear operator is closable, then its adjoint operator is densely defined and the bi-adjoint is well-defined. In the Hilbert space setting, however, we can say more about the bi-adjoint:

THEOREM 3.5. If the linear operator A is closable, then $A^{**} = \overline{A}$, which implies that if A is closed, then $A^{**} = A$ holds.

The concept of Hilbert adjoint gives rise to introduce the following property that a Hilbert space operator possibly owns: the densely defined linear operator $A: \mathfrak{D}(A) \subset H \to H$ is called *normal*, if $A\,A^* = A^*\,A$ in the sense of section 2.

Let us consider some special cases of operators in Hilbert space, wich are defined by means of the adjoint. A densely defined linear operator A acting in H is called

1. *symmetric* if $A \prec A^*$;
2. *self-adjoint* if $A = A^*$;
3. *skew-adjoint* if $A = -A^*$;
4. *unitary*, if $\mathfrak{D}(A) = H$, A is bijective and $A^{-1} = A^*$.

It should be clear that every self-adjoint or skew-adjoint or unitary operator is normal. Symmetric and self-adjoint operators and their spectral theory will be discussed more in detail in sections 7 and 8.

4. Resolvent and Spectrum

Spectral theory is a subject within functional analysis that proves to be helpful in the study of abstract linear equations of type $Ax = y$ as well as of eigenvalue equations of type $Ax - \lambda x = y$.

In spectral theory one is working with complex Banach spaces. But notice that each real Banach space X can be made into a complex Banach space $X_{\mathbb{C}}$, where the related procedure is denoted the *complexification* of X: let $X_{\mathbb{C}} := X + iX = \{x + iy : x, y \in X\}$ and furnished with the addition and multiplication derived from those in X. Then the norm in $X_{\mathbb{C}}$ is defined according to

$$\|x + iy\| \;:=\; \sup\{\|x\| \cos\theta + \|y\| \sin\theta : \theta \in [0, 2\pi)\},$$

in which the space $X_{\mathbb{C}}$ proves to be complete.

Let $\mathbf{1}_X$ be the identity mapping in a Banach space X. Given a linear operator $A : \mathfrak{D}(A) \subset X \to X$, the basic idea in spectral theory is to split the complex plane $\mathbb{C}$ into two disjoint sets $\rho(A)$ and $\sigma(A)$, denoted the *resolvent set* and the *spectrum* of A, respectively, where by definition $\zeta \in \rho(A)$ if and only if

(R-1) $A - \zeta := A - \zeta\,\mathbf{1}_X : \mathfrak{D}(A) \to X$ is bjective,
(R-2) the inverse mapping $(A - \zeta)^{-1} : X \to \mathfrak{D}(A)$ is continuous,

otherwise ζ is said to belong to $\sigma(A)$. The numbers in the sets $\rho(A)$ and $\sigma(A)$ are called *regular values* and *spectral values*, respectively. If A is closed, however, property (R-2) is a consequence of the closed graph theorem and does not have to be postulated by its own.

The resolvent set $\rho(A)$ may be the empty set, a proper subset of the complex plane or the set $\mathbb{C}$ itself. We summarize a couple of further properties of the resolvent set in the following

THEOREM 4.1. Let A be a linear operator acting in X. Then:
(a) $\rho(A)$ is open.
(b) If $\rho(A) \neq \emptyset$, then A is closed (and often denoted *regular*).
(c) If A is continuous and everywhere defined in X, then we actually have $\emptyset \neq \rho(A) \neq \mathbb{C}$ and $|\lambda| \leq \|A\|$ for all $\lambda \in \sigma(A)$.

If X is a Banach space and A is assumed to be regular, the operator-valued mapping $R(A, \cdot) : \rho(A) \to \mathscr{L}(X)$ defined by

$$\zeta \mapsto R(A, \zeta) \;:=\; (A - \zeta)^{-1}$$

is called the *resolvent function* or briefly the *resolvent* of A.

Remember the concept of a vector-valued holomorphic function: if $G \subset \mathbb{C}$ is an open domain, a function $f : G \to X$ is called *holomorphic* if for every $\phi \in X^*$ the composition $\phi \circ f : G \to \mathbb{C}$ is holomorphic in the sense of complex analysis. The function f is holomorphic if and only if f is *analytic*, that is, if $f(\zeta)$ for ζ close to $\zeta_0 \in \mathcal{O}$ can locally be represented by a power series

$$f(\zeta) = \sum_{k=0}^{\infty} a_k(\zeta_0)\,(\zeta - \zeta_0)^k \quad \text{for } |\zeta - \zeta_0| < R(\zeta_0).$$

THEOREM 4.2. The resolvent is holomorphic in every $\zeta_0 \in \rho(A)$, and $R(A,\zeta)$ can be represented by a power series

$$\sum_{n \in \mathbb{N}} b_n(\zeta_0)\,(\zeta - \zeta_0)^n$$

for $|\zeta - \zeta_0|$ sufficiently small and with coefficents $b_n(\zeta_0) \in \mathscr{L}(X)$.

All the mappings $R(A,\zeta)$ are coupled by the *resolvent equation*

$$R(A,\zeta_1) - R(A,\zeta_2) \;=\; (\zeta_2 - \zeta_1)\,R(A,\zeta_1) \circ R(A,\zeta_2) \tag{4.1}$$

for $\zeta_1, \zeta_2 \in \rho(A)$, which especially implies that the linear operators $R(A,\zeta_1)$ and $R(A,\zeta_2)$ commute with each another.

Estimates for the operator norm of resolvents play an important role in later sections. In fact, the estimate

$$\|R(A,\zeta)\| \;\geq\; r(R(A,\zeta)) \;\geq\; \operatorname{dist}(\zeta,\sigma(A))^{-1} \tag{4.2}$$

holds for any $\zeta \in \rho(A)$, where $r(R(A,\zeta)) = \sup\{|\lambda| : \lambda \in \sigma(R(A,\zeta))\}$ is the *spectral radius* of $R(A,\zeta)$ and $\operatorname{dist}(\zeta,\sigma(A))$ is the minimal distance between the singleton $\{\zeta\}$ and the spectrum $\sigma(A)$.

The estimate (4.2) is a consequence of the so-called *spectral mapping theorem*, establishing a connection between the spectra of the resolvents $R(A,\zeta)$ and the operator A itself and reads as follows:

THEOREM 4.3. Let A be a linear operator with spectrum $\sigma(A)$ and resolvent function $R(A,\cdot)$. Then for $\zeta \in \rho(A)$ we have

$$\sigma(R(A,\zeta)) \setminus \{0\} \;=\; \{(\lambda - \zeta)^{-1} : \lambda \in \sigma(A)\}.$$

We next will have a look at the spectral values of a closed linear operator A acting in a Banach space X and study for $\lambda \in \sigma(A)$ the possible reasons that prevent the operator $A-\lambda$ to be bijective.

1. Recall from linear algebra that a complex number λ is called an *eigenvalue* of a linear operator A, if there is an element $x \in X$ such that $x \neq 0$ and $A\,x = \lambda x$. In this case the element x is called an *eigenvector* of A with respect to λ, and all these eigenvectors form a linear subspace of X, called the *eigenspace* of A for λ.

   The *point spectrum* of A is defined to be the set $\sigma_p(A)$ of all eigenvalues of A. Clearly, for $\lambda \in \sigma_p(A)$ the operator $A-\lambda$ is not injective, since its kernel is non-trivial.

If $A-\lambda$ is injective, but not surjective, then $(A-\lambda)^{-1}$ is defined on the range $\mathfrak{R}(A-\lambda)$. We distinguish between two cases:

2. if $\mathfrak{R}(A-\lambda)$ is dense in X, then λ is said to be an element of the set $\sigma_c(A)$, called the *continuous spectrum* of A, otherwise
3. the number λ belongs to the *residual spectrum* $\sigma_r(A)$.

In summary, for each closed linear operator $A : \mathfrak{D}(A) \subset X \to X$ the set of complex numbers forms a disjoint union of the resolvent set and the three different kinds of spectra, namely

$$\mathbb{C} \;=\; \rho(A) \sqcup \sigma_p(A) \sqcup \sigma_c(A) \sqcup \sigma_r(A).$$

The decomposition of the spectrum into these three parts is the classical approach.

The *approximate point spectrum* $\sigma_{ap}(A)$ of a closed linear operator A is the set of $\lambda \in \mathbb{C}$, for which $A-\lambda$ is not injective or $\mathfrak{R}(A-\lambda)$ is not closed in X. The elements of $\sigma_{ap}(A)$ are called the *approximate eigenvalues* of A. Any number $\lambda \in \mathbb{C}$ belongs to $\sigma_{ap}(A)$ if and only if there is a sequence $\{x_n\}_{n\in\mathbb{N}} \subset \mathfrak{D}(A)$ such that $\|x_n\| = 1$ for all $n \in \mathbb{N}$ and $\lim_{n\to\infty} \|A\,x_n - \lambda x_n\| = 0$. Since A is closed, the topological boundary $\partial\sigma(A)$ of the spectrum forms a subset of $\sigma_{ap}(A)$.

It is easy to see that each eigenvalue of A belongs to $\sigma_{ap}(A)$, so this set generalizes the point spectrum $\sigma_p(A)$. But $\sigma_{ap}(A)$ can be empty only in the case, where $\sigma(A) = \emptyset$ or $\sigma(A) = \mathbb{C}$.

Up to now we did not assume that the linear operators under consideration are densely defined. But if this is true for a linear operator A, then the adjoint A^* is uniquely defined, and one may ask for relations between the spectra of A and A^*. In fact, it turns out that the resolvent set $\rho(A^*)$ and spectrum $\sigma(A^*)$ are mirrored sets of $\rho(A)$ and $\sigma(A)$, and each one can be obtained by the other one by reflection at the real line. There is also an interesting identity between the residual spectrum of A and the point spectrum of A^*, namely $\sigma_r(A) = \sigma_p(A^*)$ (see [5], p. 161).

If a linear operator A is bounded and defined in the entire space X, we have $\sigma(A) \neq \emptyset$, which is a consequence of LIOUVILLE's theorem in complex analysis, quoting that a bounded and holomorphic function $f : \mathbb{C} \to \mathbb{C}$ must be constant.

Furthermore, the *spectral radius* of A is defined according to

$$r(A) := \sup\{|\lambda| : \lambda \in \sigma(A)\}.$$

It can be shown that $r(A) \leq \|A\|$, and since each resolvent $R(A,\zeta)$ is bounded, we have the estimation

$$\|R(A,\zeta)\| \geq r(R(A,\zeta)) \geq \delta(\zeta,\sigma(A))^{-1} \tag{4.3}$$

for any $\zeta \in \rho(A)$, where $\delta(\zeta,\sigma(A))$ is the minimal distance between the singleton $\{\zeta\}$ and the spectrum $\sigma(A)$. Notice that inequality (4.3) is a consequence of the spectral mapping theorem.

A further main subject in this section is the exposition of the *holomorphic functional calculus*, sometimes also called the RIESZ-DUNFORD *calculus* for bounded linear operators. It allows to define functions $f(A)$ for a given linear operator A.

To introduce this kind of functional calculus, let $H(G)$ be the space of analytic functions in G. We denote $H^\infty(G)$ the subspace of all *bounded* functions in $H(G)$, which is a Banach space with respect to the norm

$$\|f\|_\infty := \sup_{\zeta \in G} |f(\zeta)|.$$

Let $G \subset \mathbb{C}$ be an open and connected subset of the complex plane, $g : [a,b] \to G$ be a rectifiable path and $f : G \to \mathbb{C}$ be a complex-valued function. Then

$$\int_\gamma f(\zeta)\, d\zeta := \int_a^b f\, dg$$

is called the *contour integral* of f with respect to the function g, where γ is the image of $[a,b]$ under g.

CAUCHY*'s integral theorem* is a corner stone in complex analysis and quotes that if f is analytic in G, then for *every* closed curve γ we have

$$\int_\gamma f(\zeta)\, d\xi = 0.$$

Furthermore, CAUCHY*'s integral formula* says that for each $z_0 \in G$ belonging to the interior of a closed curve γ the value $f(z_0)$ is nothing but

$$f(z_0) = \frac{1}{2\pi i} \int_\gamma \frac{f(z)}{z - z_0} \, dz \,, \tag{4.4}$$

where the *winding number* of γ has value 1. This representation of $f(z_0)$ does not depend on the shape of γ.

If we consider $z_0 \in \mathbb{C}$ as a linear operator $Z_0 : \mathbb{C} \to \mathbb{C} \in \mathscr{L}(\mathbb{C})$ defined by $Z_0 \, \zeta := z_0 \cdot \zeta$ for $\zeta \in \mathbb{C}$, then the spectrum of $\mathbb{Z}_0$ consists of the single point z_0 and the resolvent adopts the form $R(Z_0, \zeta) = (\zeta - z_0)^{-1}$. Using (4.4), we obtain

$$f(z_0) = \frac{1}{2\pi i} \int_\gamma f(\zeta) \, R(Z_0, \zeta) \, d\zeta. \tag{4.5}$$

The formula (4.5) serves as a pattern for the *holomorphic* or *analytic functional calculus* in unital and complex Banach algebras like the operator algebra $\mathscr{L}(X)$. Let A be a bounded linear operator acting in X (so the spectrum $\sigma(A)$ is non-emtpy and bounded), γ be a closed and rectificable path with winding number 1 containing $\sigma(A)$ and $f : G \to \mathbb{C}$ be a bounded and analytic function defined on an open set $G \supset \sigma(A)$. Then the linear operator $f(A)$ is defined by the *Riesz-Dunford integral*

$$f(A) := \frac{1}{2\pi i} \int_\gamma f(\zeta) R(A, \zeta) \, d\zeta. \tag{4.6}$$

The right hand side is a vector-valued Riemann-Stieltjes integral, and convergence of the intermediate sums to the element $f(A)$ is understood in the sense of the norm topology in $\mathscr{L}(X)$. The value $f(A)$ is well-defined, since f is bounded and analytic.

Recall that the set $H^\infty(G)$ of bounded and analytic functions $f : \mathcal{O} \to \mathbb{C}$ defined in an open neighbourhood $\mathcal{O}$ of $\sigma(A)$ forms a Banach algebra. Thus the mapping

$$\begin{cases} \Phi_A : H^\infty(\mathcal{O}) \to \mathscr{L}(B), \\ f \mapsto f(A) \text{ as defined in (4.6)} \end{cases}$$

forms a *Banach algebra homomorphism* from $H^\infty(\mathcal{O})$ to $\mathscr{L}(B)$, that is, the mapping Φ_A is linear, norm-preserving and respects the multiplicative operations in both algebras. But Φ_A also maps polyomials $p \in H^\infty(\mathcal{O})$ to polynomials $p(A) \in \mathscr{L}(B)$. Due to these properties, Φ_A is called a *functional calculus*, which in addition deserves the appendix "analytic", since the space $H^\infty(\mathcal{O})$ consists of analytic functions.

It should be mentioned that for this kind of functional calculus the *spectral mapping theorem* is valid, that is, $\sigma(f(A)) = f(\sigma(A))$ for $A \in \mathscr{L}(B)$ and $f \in H^\infty(\mathcal{O})$. The resulting operator $f(A) \in \mathscr{L}(X)$ defined by (4.6) is called the RIESZ-DUNFORD *operator* associated with A and $f \in H^\infty(G)$.

In the following we suppose that λ is an *isolated* point of the spectrum $\sigma(A)$, so $\sigma(A) \cap \mathcal{U}_\lambda = \{\lambda\}$ holds for every sufficiently small neighbourhood $\mathcal{U}_\lambda$ of λ. Let γ_λ be a closed and rectifiable path, which surrounds only λ, but not any other element of $\sigma(A)$. Then for $x \in X$ the contour integral

$$\pi_\lambda(x) := \frac{1}{2\pi i} \int_{\gamma_\lambda} R(A,\zeta)x \, d\zeta$$

gives rise to a further linear mapping π_λ, called the RIESZ *projection* of A with respect to the spectral value λ.

To the end, we collect the properties of the mapping π_λ to the following list:

1. π_λ is linear.
2. π_λ owns the *projection property* $\pi_\lambda \circ \pi_\lambda = \pi_\lambda$.
3. The spectrum of this mapping only consists of the value λ_0 itself.
4. π_λ commutes with A, so $A \circ \pi_\lambda = \pi_\lambda \circ A$ holds.
5. The range Π_λ of π_λ is a closed linear subspace of X.
6. There is a closed subspace $\Pi_\lambda^\perp \subset X$ such that $X = \Pi_\lambda \oplus \Pi_\lambda^\perp$, where $\Pi_\lambda^\perp$ is well-defined by the decomposition theorem.
7. Both subspaces Π_λ and $\Pi_\lambda^\perp$ are A-invariant, so $A\,\Pi_\lambda \subset \Pi_\lambda$ and $A\,\Pi_\lambda^\perp \subset \Pi_\lambda^\perp$. One also says that the subspaces Π_λ and $\Pi_\lambda^\perp$ *reduce* the entire space X.
8. The mapping $\pi_\lambda^\perp := 1_X - \pi_\lambda$ is a projection again, and both projections π_λ and $\pi_\lambda^\perp$ are *orthogonal* to each another, that is, $\pi_\lambda \circ \pi_\lambda^\perp = \pi_\lambda^\perp \circ \pi_\lambda = 0$.

The concept of Riesz projection can be extended into different directions. First, the linear operator A may be assumed to be closed instead of being continuous, and second the subset $\{\lambda\} \subset \sigma(A)$ may be replaced by any connected component $\sigma_0(A)$ of the set $\sigma(A)$.

5. Normally Solvable Operators

The general subject in this section is the review of linear operators having closed range. In fact, given a densely defined linear operator A acting between Banach spaces X and Y, the orthogonality relations (3.2) mentioned in section 3 can be improved if the ranges of A and A^* are *closed* subspaces of Y and X^*, so in these cases we actually would have

$$\mathfrak{R}(A) = {}^{\perp}\mathfrak{N}(A^*) \quad \text{and} \quad \mathfrak{R}(A^*) = \mathfrak{N}(A)^{\perp}.$$

One also says that a linear operator A is *normally solvable* if it is densely defined and $\mathfrak{R}(A) = {}^{\perp}\mathfrak{N}(A^*)$ holds, which means by our next theorem that A has closed range.

The *closed range theorem* is one of the classical results in functional analysis and provides equivalent conditions for a densely defined and closed linear operator to own closed range:

THEOREM 5.1. Let X, Y be Banach spaces and $A : \mathfrak{D}(A) \subset X \to Y$ be densely defined and closed. Then it is equivalent to quote:
(a) $\mathfrak{R}(A)$ is closed.
(b) $\mathfrak{R}(A^*)$ is closed.
(c) $\mathfrak{R}(A)$ is the pre-annihilator of $\mathfrak{N}(A^*)$, i. e. $\mathfrak{R}(A) = {}^{\perp}\mathfrak{N}(A^*)$.
(d) $\mathfrak{R}(A^*)$ is the annihilator of $\mathfrak{N}(A)$, i. e. $\mathfrak{R}(A^*) = \mathfrak{N}(A)^{\perp}$.

For a proof of this theorem see for example [17], pp. 205-207. The statement was originally proved by S. BANACH for continuous linear operators and have later been extended to closed linear operators by F. BROWDER, see [2], theorem 1.1 and theorem 1.2.

It is natural to search for further conditions that imply normal solvability of operators, and the first of them is what is usually called the *bounded-below theorem.*

Let X and Y be Banach spaces. A linear operator A with domain of definition $\mathfrak{D}(A) \subset X$ and range space Y is called *bounded below* if there is $\theta > 0$ such that

$$\|Ax\|_Y \geq \theta \, \|x\|_X \text{ for all } x \in \mathfrak{D}(A). \tag{5.1}$$

The *bounded-below theorem* provides a fundamental assertion in operator theory and is helpful to verify the closeness of $\mathfrak{R}(A)$:

THEOREM 5.2. If A fulfills (5.1), then A is injective and the operator

$A^{-1}: \mathfrak{R}(A) \subset Y \to X$ is bounded with operator norm $\|A^{-1}\| \leq \theta^{-1}$. Furthermore, if A is closed, then $\mathfrak{R}(A)$ is a closed set in Y.

Indeed, injectivity of the operator A can easily proved, supposed (5.1) holds, and closeness of $\mathfrak{R}(A)$ is a consequence of sequentially closeness of A.

We want to extend our considerations to a more general class of linear operators. Preliminary, let $\alpha(A)$ (the *nullity index* of A) and $\beta(A)$ (the *deficiency index* of A) be the Hamel dimensions of the linear subspaces $\mathfrak{N}(A) \subset X$ and $\mathfrak{C}(A) \subset Y$ of a given linear operator $A: \mathfrak{D}(A) \subset X \to Y$.

Both integers together can be used to measure the deviation of A from bijectivity. Moreover, if at least one of them is finite, then the so-called *Fredholm index*

$$\chi(A) := \begin{cases} +\infty, & \alpha(A) = \infty \\ \alpha(A) - \beta(A), & \alpha(A) < \infty, \beta(A) < \infty \\ -\infty, & \beta(A) = \infty \end{cases}$$

is well-defined and takes values in the set $\{-\infty\} \cup \mathbb{N} \cup \{+\infty\}$. In the special case, where $\alpha(A) < \infty$, $\beta(A) < \infty$ and consequently $\chi(A) \in \mathbb{Z}$, the operator A is said to have *finite index*.

The linear operator $A: \mathfrak{D}(A) \subset X \to Y$ is called *semi-Fredholm*, if

(F-1) the range $\mathfrak{R}(A)$ is closed,

(F-2) the kernel $\mathfrak{N}(A)$ or the cokernel $\mathfrak{C}(A)$ is finite-dimensional (rsp. $\alpha(A) < \infty$ or $\beta(A) < \infty$).

Notice the remarkable fact that if A is continuous with $\mathfrak{D}(A) = X$ and $\beta(A) < \infty$, then the range $\mathfrak{R}(A)$ is necessarily closed, since it constitutes the topological complement of the finite-dimensional linear subspace $\mathfrak{C}(A) \subset Y$, that is, $X = \mathfrak{R}(A) \oplus \mathfrak{C}(A)$.

We refine the concept of semi-Fredholm operator as follows: a linear operator $A: \mathfrak{D}(A) \subset X \to Y$ with closed range $\mathfrak{R}(A)$ is called

(F-3) *upper semi-Fredholm*, if $\alpha(A) < \infty$ rsp. $\chi(A) < \infty$,

(F-4) *lower semi-Fredholm*, if $\beta(A) < \infty$ rsp. $\chi(A) > -\infty$.

Examples of upper semi-Fredholm operators are the bounded-below ones. In fact, theorem 5.2 implies that these operators are upper semi-Fredholm with vanishing nullity index.

If a continuous linear operator A satisfies an abstract *Schauder estimate*, one may quote a sufficient condition for A to be upper semi-Fredholm, which is known as *Peetre's lemma*:

THEOREM 5.3. Let X, Y, Z be three Banach spaces, where X is compactly embedded into Z, that is, there is an injective and compact linear mapping $\iota : X \to Z$. Then a continuous operator $A : X \to Y$ is upper semi-Fredholm iff there is $c > 0$ such that

$$\|x\|_X \leq c\,(\|\iota x\|_Z + \|A\,x\|_Y) \qquad (x \in X).$$

For a proof of this theorem, see [16], theorem 12.12, page 180.

Nowadays there is a well-elaborated *perturbation theory* for semi-Fredholm operators available, where conditions are studied, under which a linear operator of the form $A+B$ is semi-Fredholm, assumed A is semi-Fredholm and B fulfills certain properties. In this case the operator B is regarded as a perturbation of A.

We next quote two theorems with respect to conditions on B, which ensure the semi-Fredholm property of the operator $A+B$. We basically assume that X and Y are Banach spaces and first quote *Yood's theorem*:

THEOREM 5.4. Let $A : X \to Y$ be semi-Fredholm and $B : X \to Y$ be linear and bounded with sufficiently small norm $\|B\|$. Then $A+B$ remains semi-Fredholm with $\chi(A+B) = \chi(A)$.

In other words, this theorem asserts that the set of semi-Fredholm operators is open in the set $\mathscr{L}(X, Y)$. Especially linear operators of the form $A+\lambda$ for $|\lambda|$ small enough are semi-Fredholm again. But more can be said on the values $\alpha(A+\lambda)$, $\beta(A+\lambda)$ and $\chi(A+\lambda)$ by GOHBERG*'s punctured neighbourhood theorem*:

THEOREM 5.5. Let the continuous linear operator $A : X \to X$ be semi-Fredholm. Then there is $\lambda_0 > 0$ such that for all $\lambda \in (0, \lambda_0)$ we have $\alpha(A+\lambda) \leq \alpha(A)$, $\beta(A+\lambda) \leq \beta(A)$ and $\chi(A+\lambda) = \chi(A)$.

In the following we consider the case, where both values are finite, so we refine the concept of semi-Fredholm operator to a stronger one: in addition to (F-1) to (F-4), a continuous linear operator $A : X \to Y$ acting between normed linear spaces X and Y is called

(F-5) *Fredholm*, if $\chi(A)$ is finite, i. e. $\alpha(A) < \infty$ and $\beta(A) < \infty$.

The relationships (3.2) imply that if A is densely defined and Fredholm, then A^* is also Fredholm and $\chi(A^*) = -\chi(A)$ holds.

Sure, if A is at once self-adjoint and Fredholm, we have $\chi(A) = 0$.

We have introduced the Fredholm property of linear operators under the assumption that these are continuous and everywhere defined in their domain spaces. If a linear operator A is closed instead of continuous, however, we define the possible Fredholm property by regarding A as a continuous linear operator $A : \mathfrak{D}(A) \to Y$, where the domain of definition $\mathfrak{D}(A)$ is endowed with the graph norm. Indeed, the pair $(\mathfrak{D}(A), \|\cdot\|_A)$ constitutes a Banach space due since A is assumed to be closed.

We first collect a couple of properties of Fredholm operators:

1. A^* is Fredholm if and only if A is so.
2. If one of A or A^* is Fredholm, then the dimensions of the kernels and ranges of A and A^* are related to each other by $\alpha(A^*) = \beta(A)$ and $\beta(A^*) = \alpha(A)$, which implies $\chi(A^*) = -\chi(A)$.
3. Fredholm operators acting in X and belonging to one and the same connected component within the algebra $\mathscr{L}(X)$ have identical index. Thus, if A is Fredholm and the linear operator B can be reached by a continuous path in $\mathscr{L}(X)$ with starting point A, then B is Fredholm, too, and $\chi(B) = \chi(A)$ holds.

Our next subject is the representation of Fredholm operators using compact linear operators. More precisely, a Fredholm operator can be represented by means of a *left* and *right parametrix*, respectively.

Let $A : X \to Y$ be a continuous linear operator. Then a linear mapping $P_l : Y \to X$ is called a *left parametrix* of A if $A \circ P_l = \mathbf{1}_X + k_l$ for some compact operator $k_l \in \mathscr{L}(X)$. In a similar way, a linear mapping $P_r : Y \to Y$ is said to be a *right parametrix* of A, if $P_r \circ A = \mathbf{1}_Y + k_r$ for some compact operator $k_r \in \mathscr{L}(Y)$.

A linear operator A is Fredholm if and only if it owns a left and right parametrix, where the involved compact operators k_l and k_r are even of *finite-rank*, that is, they have finite-dimensional ranges.

In the following we cite the *Fredholm alternative theorem* as the outstanding result in Fredholm theory with respect to the solvability of the linear equation $Ax = y$. Remember the *rank-nullity theorem* known from linear algebra, claiming that if a linear space V is finite-dimensional, then a linear mapping $A : V \to V$ is surjective if and only if it is injective.

The alternative theorem is a considerable generalization of the rank-nullity theorem to arbitrary Banach spaces and Fredholm operators with vanishing index:

THEOREM 5.6. Let X, Y be Banach spaces and the continuous linear operator $A: X \to Y$ be Fredholm with index $\chi(A) = 0$.

According to the solvability of the linear equation $A\,x = y$, exactly one of the following cases occurs:

(a) $A\,x = y$ is uniquely solvable for all $y \in X$, so A is bijective,

(b) $A\,x = y$ is not for all $y \in X$ solvable, and A is neither injective nor surjective in this case. However, if the equation is solvable for an element $y_0 \in X$, then the set of solutions forms a finite-dimensional subspace of X. Moreover, the equation $A\,x = y$ owns a solution $x \in X$ if and only if $y \in {}^{\perp}\mathfrak{N}(A^*)$.

The various concepts of semi-Fredholm operators can be used to characterize the *essential spectrum* $\sigma_{ess,0}(A)$ of a closed operator A acting in a complex Banach space X: a number $\lambda \in \mathbb{C}$ belongs to $\sigma_{ess,0}(A)$ if and only if $A - \lambda$ is *not* semi-Fredholm. But there are further approaches to define the notion of essential spectrum:

$\sigma_{ess,1}(A) := \{\lambda \in \mathbb{C} : A - \lambda \text{ is not upper semi-Fredholm}\}$,
$\sigma_{ess,2}(A) := \{\lambda \in \mathbb{C} : A - \lambda \text{ is not lower semi-Fredholm}\}$,
$\sigma_{ess,3}(A) := \{\lambda \in \mathbb{C} : A - \lambda \text{ is not emi-Fredholm}\}$,
$\sigma_{ess,4}(A) := \{\lambda \in \mathbb{C} : A - \lambda \text{ is Fredholm}\}$,
$\sigma_{ess,5}(A) := \{\lambda \in \mathbb{C} : A - \lambda \text{ is not Fredholm with } \chi(A - \lambda) = 0\}$,
$\sigma_{ess,6}(A) := \{\lambda \in \sigma_{ess,5}(A) : \lambda \text{ is an isolated point in } \sigma(A)\}$.

The sets $\sigma_{ess,4}(A)$ and $\sigma_{ess,5}(A)$ are called the *Wolf spectrum* and the *Schechter spectrum*, respectively, whereas the set $\sigma_{ess,6}(A)$ is denoted the *Browder spectrum*.

All sets $\sigma_{ess,k}(A)$ are closed subsets of $\sigma(A)$, and by definition we have the chain of inclusions

$$\sigma_{ess,3}(A) \subset \sigma_{ess,1}(A) \cup \sigma_{ess,2}(A) \subset \sigma_{ess,4}(A) \subset \sigma_{ess,5}(A) \subset \sigma_{ess,6}(A).$$

In the Hilbert space setting, however all sets $\sigma_{ess,k}(A)$ coincide and form the largest subset $\sigma_{ess}(A) \subset \sigma(A)$ such that for every compact linear operator $k \in \mathscr{L}(X)$

$$\sigma_{ess}(A + k) \;=\; \sigma_{ess}(A).$$

6. Symmetry and Self-Adjointness

This section serves as an overview on symmetric and self-adjoint linear operators acting in a Hilbert space. Let us begin with some remarks on the numerical range.

Besides the spectrum $\sigma(A)$, the *numerical range* $\Theta(A)$ plays an important role in the general theory of a Hilbert space operator A and is defined by

$$\Theta(A) := \{\langle A, x\rangle : x \in \mathfrak{D}(A), \|x\| = 1\}.$$

It is a well-known fact, given by the *Toeplitz-Hausdorff theorem*, that $\Theta(A)$ is a convex set within the complex plane. There are relationships between the numerical range and the some kinds of spectra of the operator A. Indeed, we have $\sigma_p(A) \subset \Theta(A)$, $\sigma_{ap}(A) \subset \overline{\Theta(A)}$ and even $\sigma(A) \subset \Theta(A)$ for $A \in \mathscr{L}(H)$, (see [8], theorem A.23 on p. 266).

Let $\Delta(A) := \mathbb{C} \setminus \overline{\Theta(A)}$ be the *external field of regularity* of A. If $\Delta(A) \neq \emptyset$, then it consists of one or two connected components. More precisely, if $\Delta(A)$ has two connected components, then the numerical range is either a strip or a ray, otherwise it is a half-strip or a half-ray.

For $\zeta \in \Delta(A)$ we of course have $\langle Ax, x\rangle \neq \zeta$ for $x \in \mathfrak{D}(A)$ and $\|x\| = 1$, and the following chain of inequalities

$$0 < \operatorname{dist}(\zeta, A) \leq |\langle Ax, x\rangle - \zeta| = |\langle (A-\zeta)x, x\rangle| \leq \|(A-\zeta)x\|$$

is valid, where $\operatorname{dist}(\zeta, A)$ denotes the distance between the singleton $\{\zeta\} \subset \Delta(A)$ and the set $\overline{\Theta(A)}$, which in fact is positive, since these sets are closed and have empty intersection. This implies that for $\zeta \in \rho(A) \cap \Delta(A)$ we have

$$\|R(A, \zeta)\| \leq \frac{1}{\operatorname{dist}(\zeta, \Theta(A))}.$$

For $\zeta \in \Delta(A)$ the operator $A-\zeta$ is bounded below and $\Delta(A)$ is a subset of the *field of regularity* $\Pi(A) := \{\zeta \in \mathbb{C} : A-\zeta \text{ is bounded below}\}$.

A linear operator A acting in H and with domain of definition $\mathfrak{D}(A)$ dense in H is called *symmetric*, if the adjoint A^* forms an extension of A, and this property is equivalent to

$$\langle Ax, y\rangle = \langle x, Ay\rangle \quad \text{for all } x, y \in \mathfrak{D}(A).$$

But if $\mathfrak{D}(A)$ is *not* dense in H, it is still possible to define potential symmetry of A: in this case, A is called *symmetric*, if $\Theta(A) \subset \mathbb{R}$. It can be shown that if A is densely defined, then the both mentioned concepts of symmetry are equivalent.

Finally, a symmetric linear operator A acting in H is called *half-bounded*, if $\Theta(A) \subset (-\infty, a)$ or $\Theta(A) \subset (b, \infty)$ holds.

Densely defined and symmetric linear operators distinguish by a couple of properties. Indeed, if A is symmetric, then

(a) A is closable, since A^* forms a closed extension of A;
(b) the closure $\overline{A}$ is symmetric, too;
(c) all eigenvalues of A are real;
(d) eigenvectors for different eigenvalues of A are *orthogonal* to each other.

A useful property of symmetric closed operators is given by the fact that the operators $A-\zeta$ are bounded below for $\operatorname{Im}\zeta \neq 0$, hence they are injective and own closed ranges, so they are semi-Fredholm operators with $\alpha(A-\zeta)=0$. Furthermore, the codimensions $\beta(A-\zeta)$ of the ranges $\mathfrak{R}(A-\zeta)$ are constant in the upper and lower half-plane of $\mathbb{C}$, respectively, which is a consequence of the stability theorem for semi-Fredholm operators. The values $d_\pm := d(\pm i) \in \mathbb{N}_0 \cup \{\infty\}$ are called the *defect indices* of A.

Now the decisional question is whether the operators $A \pm \zeta$ are surjective and the defect indices $d_\pm$ of A are zero, respectively. To the end we can distinguish four possible cases:

1. $d_+ \neq 0$ and $d_- \neq 0$. The operators $A+i$ and $A-i$ are not surjective, hence the upper and lower complex half-plane belong to the spectrum $\sigma(A)$. Because the spectrum is closed, the real line also belongs to $\sigma(A)$, and in conclusion we have $\sigma(A)=\mathbb{C}$. In addition, for $\operatorname{Im}\zeta \neq 0$ we have $\zeta \in \sigma_r(A)$.
2. $d_+ = 0$ and $d_- \neq 0$. The operator $A+i$ is surjective, but the operator $A-i$ is not so. Hence the lower complex half-plane and the real line belong to $\sigma(A)$, whereas the upper complex half-plane belongs to $\rho(A)$.
3. $d_+ \neq 0$ and $d_- = 0$. The operator $A-i$ is surjective, but the operator $A+i$ is not so. Hence the upper complex half-plane and the real line belong to $\sigma(A)$, whereas the lower complex half-plane belongs to $\rho(A)$.
4. $d_+ = 0$ and $d_- = 0$. The operators $A+i$ and $A-i$ are surjective, hence the upper and lower complex half-plane belong to the resolvent set $\rho(A)$. The spectrum $\sigma(A)$ forms a subset of the real

line in this case.

It should be notified that the adjoint A^* of a symmetric linear operator A is not necessarily symmetric again. If A is symmetric, then every other linear operator S having the property $A \prec S$ is called a *symmetric extension* of A. Notice that $A \prec S$ always implies $S^* \prec A^*$, which is also true for non-symmetric operators A and S.

Now, let $S_1, \ldots, S_k$ be a finite sequence of symmetric extensions of A, where S_{i+1} extends S_i for $i = 1, \ldots, k-1$. Since $S_k \prec S_k^*$ and $S_{i+1}^* \prec S_i^*$ for $i = 1, \ldots, k-1$, we obtain by concatenating all these extensions the chain

$$A \prec S_1 \prec S_2 \prec \ldots \prec S_k \prec S_k^* \prec \ldots \prec S_2^* \prec S_1^* \prec A^*.$$

Thus one may hope to find a *maximal* symmetric extension S_{sa} of A such that

$$A \prec S_{sa} = S_{sa}^* \prec A^*$$

holds, and in this case the operator S_{sa} cannot be extended to a larger symmetric operator. This expectation suggests to introduce the notion of *self-adjointness*: a symmetric linear operator A is called *self-adjoint*, if $A^* \prec A$, or equivalently, if $A = A^*$.

Let us mention that if a self-adjoint linear operator A is Fredholm, then $\chi(A) = 0$, which follows from the equality $\chi(A) = -\chi(A^*)$. Remember from section 5 that the various definitions of the essential spectrum of A are equivalent for any linear operator A acting in a Hilbert space. Hence we have

$$\sigma_{ess,1}(A) = \sigma_{ess,2}(A) = \sigma_{ess,3}(A) = \sigma_{ess,4}(A) = \sigma_{ess,5}(A) = \sigma_{ess,6}(A),$$

which gives rise to define $\sigma_{ess}(A) := \sigma_{ess,k}$ for any $1 \leq k \leq 6$.

Recall that the discrete spectrum $\sigma_d(A) := \sigma(A) \setminus \sigma_{ess}(A)$ is nothing but the set of isolated eigenvalues of A having finite algebraic multiplicity, so $\mathfrak{N}(A-\lambda)$ is finite-dimensional for every $\lambda \in \sigma_d(A)$. On the other hand, any value λ belongs to $\sigma_{ess}(A)$, if there is a sequence $\{x_n\}_{n \in \mathbb{N}}$ of orthonormal elements in H such that $\|Ax_n - \lambda x_n\| \to 0$ for $n \to \infty$, which is denoted a *Weyl sequence* with respect to λ.

To ensure self-adjointess of a symmetric linear operator A, the *range condition* reflects the forth case in our considerations on the defect indices of A and serves as the main tool to verify this pleasant property:

THEOREM 6.1. A densely defined, closed and symmetric linear operator A acting in H is self-adjoint iff there is $\zeta \in \mathbb{C}$ such that

$$\mathfrak{R}(A+\zeta) = \mathfrak{R}(A+\overline{\zeta}) = H,$$

or equivalently, if the defect indices of $A \pm \zeta$ fulfill $d(\pm\zeta) = 0$.

If it is even possible to find $\zeta \in \mathbb{R}$ fulfilling the range condition $\mathfrak{R}(A+\zeta) = H$, then A also proves to be self-adjoint.

A symmetric linear operator is denoted *essentially self-adjoint*, if its closure proves to be self-adjoint. There is a comparable criterion for essentially self-adjointness available:

THEOREM 6.2. A densely defined and symmetric linear operator A acting in H is essentially self-adjoint iff there is $\zeta \in \mathbb{C}$ such that

$$\overline{\mathfrak{R}(A+i)} = \overline{\mathfrak{R}(A-i)} = H.$$

Essentially self-adjointness can also be characterized by means of the adjoint: a densely defined and symmetric linear operator A is essentially self-adjoint iff the adjoint A^* is symmetric, too.

Let us now deal with operators of the form $A+B$, where A is self-adjoint and B is symmetric with domain of definition $\mathfrak{D}(B) \supset \mathfrak{D}(A)$, so $A+B$ is well-defined with $\mathfrak{D}(A+B) := \mathfrak{D}(A)$. It is our aim to find conditions on the operator B to obtain self-adjointness of $A+B$.

The following concept proves to be useful: a symmetric linear operator $B : \mathfrak{D}(B) \subset H \to H$ with $\mathfrak{D}(A) \subset \mathfrak{D}(B)$ is called *A-bounded*, if there are constants $\theta, c \geq 0$ such that

$$\|B\,x\| \leq \theta\,\|A\,x\| + c\,\|x\| \qquad \text{for all } x \in \mathfrak{D}(A). \tag{6.1}$$

In order to ensure A-boundedness of B, it is sufficient to ensure (6.1) only for the elements in a core A_0 of A.

The infimum of all numbers θ, for which there is c such that (6.1) holds, is called the *A-bound* of B. Notice that if B is a bounded operator, then it is A-bounded with A-bound $\theta < 1$ for any A. In addition, the validness of the estimate

$$\|B\,x\|^2 \leq \theta^2\,\|A\,x\|^2 + c^2\,\|x\|^2 \qquad \text{for all } x \in \mathfrak{D}(A).$$

is a sufficient condition for (6.1) to be true.

The KATO-RELLICH *theorem* provides a criterion on self-adjointness of the operator $A+B$:

THEOREM 6.3. Let the linear operator A be self-adjoint and the linear operator B with $\mathfrak{D}(B) \supset \mathfrak{D}(A)$ be symmetric and A-bounded with $\theta < 1$. Then $A+B$ with domain of definition $\mathfrak{D}(A+B) = \mathfrak{D}(A)$ is self-adjoint.

A similar result is obtained for the case, where *essentially self-adjointness* of a linear operator A should be preserved by adding a symmetric and A-bounded perturbation B:

THEOREM 6.4. Let A be essentially self-adjoint and assume that B is symmetric and A-bounded with bound $\theta < 1$. Then the operator $A+B$ is essentially self-adjoint, where $\overline{A+B} = \overline{A} + \overline{B}$.

We finally want to mention an important class of symmetric and self-adjoint linear operators. Remember from integration theory that for a given measure space $(X, \mathcal{B}, \mu)$ the linear space $L^2(X)$ of measurable and square-integrable functions $f : X \to \mathbb{C}$ forms a Hilbert space with inner product

$$\langle f, g \rangle := \int_X f \, \overline{g} \, d\mu.$$

If $m : X \to \mathbb{C}$ is a measurable function, then the linear mapping M_m defined by

$$\begin{cases} \mathfrak{D}(M_m) & := \{ f \in L^2(X) : \ m f \in L^2(X) \}, \\ (M_m f)(x) & := m(x) \cdot f(x) \ \text{ for } x \in X \text{ and } f \in \mathfrak{D}(M_m) \end{cases}$$

is called a *multiplication operator* with *generator* m. It turns out that the mapping M_m is symmetric iff m is real-valued, and since the domains of definition of the operators M_m and M_m^* coincide, each multiplication operator with real-valued generator is self-adjoint.

7. Spectral Theory of Self-Adjoint Operators

The spectral theory of self-adjoint linear operators is one of the pearls in the mathematical area of functional analysis and has important applications in physics and engineering science with a multiplicity of useful results.

First of all, we may characterize the self-adjointness of a linear operator acting in a Hilbert space by means of its spectrum, which is an immediate consequence of the range condition 6.1:

THEOREM 7.1. A linear operator A is self-adjoint iff $\sigma(A) \subset \mathbb{R}$.

A further result is given by the *resolvent estimate*: if A is self-adjoint, then for $\operatorname{Im}\zeta \neq 0$ the resolvent estimation

$$\|R(A,\zeta)\| \leq |\operatorname{Im}\zeta|^{-1}$$

holds, which follows from the inequality $|\operatorname{Im}\zeta| \leq \operatorname{dist}(\zeta, \sigma(A))$ and the general estimate (4.2).

Moreover, if A is self-adjoint and $\lambda \in \sigma(A)$, then λ is either an eigenvalue of A or $\mathfrak{R}(A-\lambda)$ is dense in H, but not the whole of H. Hence in the first case the value λ belongs to the point spectrum $\sigma_p(A)$ of A, whereas the second case means that λ belongs to the continuous spectrum $\sigma_c(A)$. This means to say the residual spectrum $\sigma_r(A)$ is empty.

The most important theorem with respect to self-adjoint linear operators, however, is the celebrated *spectral theorem.*

Recall that a *projection* in H is a symmetric linear operator P such that $P \circ P = P$. Then a *spectral resolution* is a one-parameter family $\{P_\lambda\}_{\lambda \in \mathbb{R}}$ of linear operators $P_\lambda : H \to H$ such that

(R-1) P_λ is a projection for each $\lambda \in \mathbb{R}$;
(R-2) for each $x \in H$ the mapping $\lambda \mapsto P_\lambda x$ is right-continuous;
(R-3) $\lambda < \mu$ implies $P_\lambda \leq P_\mu$, so $\langle P_\lambda x, x\rangle \leq \langle P_\mu x, x\rangle$ for $x \in H$;
(R-4) $\lim_{\lambda \to -\infty} P_\lambda x = 0$ for all $x \in H$;
(R-5) $\lim_{\lambda \to \infty} P_\lambda x = x$ for all $x \in H$.

Notice also that each projection P_λ is related to a closed linear subspace $\mathcal{A}_\lambda \subset H$. Moreover, the difference $P_\mu - P_\lambda$ for $\mu \geq \lambda$ also constitutes a projection related to the closed subspace $\mathcal{A}_\mu \ominus \mathcal{A}_\lambda$.

A spectral resolution can be used to define an operator A in H by taking the spectral integral

$$Ax = \int_{\mathbb{R}} \lambda \, dP_\lambda x, \tag{7.1}$$

where the domain of definition $\mathfrak{D}(A)$ consists of those elements $x \in H$, for which this improper Riemann-Stieltjes integral converges. The mapping A is said to be the *spectral operator* associated with the resolution $\{P_\lambda\}_{\lambda \in \mathbb{R}}$ and proves to be linear and even self-adjoint.

The *spectral theorem* for self-adjoint operators claims that the inverse statement also holds:

THEOREM 7.2. Each self-adjoint linear operator A owns a spectral resolution $\{P_\lambda\}_{\lambda \in \mathbb{R}}$ such that

$$\mathfrak{D}(A) = \{ x \in H : \int_{\mathbb{R}} \lambda^2 \, d\langle P_\lambda x, x \rangle < \infty \}$$

and

$$\langle A x, y \rangle = \int_{\mathbb{R}} \lambda \, d\langle P_\lambda x, y \rangle \quad \text{for } x \in \mathfrak{D}(A) \text{ and } y \in H.$$

In summary, self-adjoint linear operators are in a one-to-one relationship to spectral operators defined by formula (7.1).

We next want to point out the relationship between the shape of the spectrum $\sigma(A)$ and the spectral resolution of A. The following theorem shows that the regular values and the different kinds of spectral values of A are determined by the spectral resolution:

THEOREM 7.3. Let A be self-adjoint and $\mathcal{A}_\mu$ be the closed subspace in H associated with the projection P_μ for some $\mu \in \mathbb{R}$. Then

1. $\mu \in \mathbb{R}$ is a *regular value* of A iff there is an open interval $(\alpha, \beta) \subset \mathbb{R}$ such that $\mu \in (\alpha, \beta)$ and $P_\nu = P_\mu$ for each $\nu \in (\alpha, \beta)$.
2. $\mu \in \mathbb{R}$ belongs to the *point spectrum* $\sigma_p(A)$ iff
$$0 < \dim\, (\mathcal{A}_{\mu+0} \ominus \mathcal{A}_\mu) < \infty.$$
3. $\mu \in \mathbb{R}$ belongs to the *continuous spectrum* $\sigma_c(A)$ iff for each $\epsilon > 0$ the subspace $\mathcal{A}_{\mu+\epsilon} \ominus \mathcal{A}_{\mu-\epsilon}$ is infinite-dimensional.

The spectral theorem forms the base for the so-called *functional calculus* for self-adjoint operators: if a given function $f : \mathbb{R} \to \mathbb{C}$ is sufficiently regular (for example, if f is piecewise continuous and the points of discontinuity do not have any accumulation point in the real line), then it is possible to define the operator $f(A)$ by means of the spectral resolution $\{P_\lambda\}_{\lambda \in \mathbb{R}}$ of A:

$$f(A)\, x := \int_{\mathbb{R}} f(\lambda)\, dP_\lambda x,$$

where the domain of definiton of the operator $f(A)$ is defined by

$$\mathfrak{D}(f(A)) := \{x \in H : \int_{\mathbb{R}} |f(\lambda)|^2 \, dP_\lambda x < \infty\}.$$

Notice that the operator $f(A)$ is self-adjoint iff f is real-valued.

The functional calculus for unbounded self-adjoint operators owns the following properties:

1. $f(\alpha A) = \alpha f(A)$ for $\alpha \in \mathbb{C}$.
2. $f(A) + g(A) \prec (f + g)A$.

3. $f(A)g(A) \prec f(g(A))$.
4. $f(A)^* = f(A^*)$.
5. $\sigma(f(A)) = f(\sigma(A))$ (*spectral mapping theorem*).

We are especially interested in the case, where the self-adjoint operator A is positive and $f(\lambda) := 0$ for $t < 0$ and $e^{-\lambda t}$ for $t \geq 0$, in which the family $\{e^{-tA}\}_{t\geq 0}$ of bounded linear operators is shown to constitute an analytic semigroup, see section 11). But also other operators $f(A)$ are meaningful in applications, which for example are given by the functions $x \mapsto \sqrt{x}$, $x \mapsto |x|$, $x \mapsto \cos x$, $x \mapsto \sin x$ and are denoted as $\sqrt{A}$, $|A|$, $\cos A$, $\sin A$.

Especially the resolvents $R(A,\zeta) := (A-\zeta)^{-1}$ for $\zeta \in \rho(A)$ of a self-adjoint linear operator A can be represented by means of the spectral resolution, that is

$$R(A,\zeta)\, x \;=\; \int_{-\infty}^{\infty} \frac{1}{\lambda - \zeta}\, dP_\lambda x, \quad (x \in H,\, \operatorname{Im}\zeta \neq 0).$$

On the other hand, STONE*'s formula* provides a constructive method to describe the spectral resolution of A in an interval $[a,b]$ by means of the resolvents $R(A,\zeta)$. Let $x \in H$ be arbitrary, then

$$\frac{1}{2}\{P[a,b]+P(a,b)\}\, x = \lim_{\epsilon \downarrow 0} \frac{1}{2\pi i} \int_a^b \{R(A,t+i\epsilon)\, x - R(A,t-i\epsilon)\, x\}\, dt.$$

The presented functional calculus provides applications to initial-value problems for evolution equations governed by self-adjoint and positive linear operators. In the rest of this section we shall elaborate the usage of this method.

First of all let us clarify what is meant by a differentiable vector-valued function. If X is an arbitrary Banach space and $[a,b] \subset \mathbb{R}$, then a vector-valued function $x : [a,b] \to X$ is said to be *classical differentiable* in $t_0 \in (a,b)$, if there is an element $\dot{x}(t_0) \in X$ such that

$$\lim_{t \to t_0} \left\| \frac{x(t) - x(t_0)}{t - t_0} - \dot{x}(t_0) \right\| = 0.$$

Moreover, the function x is called classical differentiable in (a,b), if x is so in each $t \in (a,b)$. The linear space consisting of functions having *continuous* classical derivatives in (a,b) is denoted $C^1(a,b;X)$.

The spaces $C^k(a,b;X)$ $(k \in \mathbb{N})$ are defined recursively as usual, and especially the space $C^\infty(a,b;X)$ is defined by taking the intersection of all spaces $C^k(a,b;X)$ for $k \in \mathbb{N}_0$.

The initial-value problems we want to consider in this section are the homogeneous Cauchy problems of first and second order and the abstract Schrödinger equation. Let us notice in advance that they can be solved as well by semigroup methods, see also the sections $9, 10$ and 11.

1. The *homogeneous Cauchy problem of first order* is the initial-value problem given by

$$\begin{cases} \dot{x}(t) + A\,x(t) = 0, \\ x(0) = x_0 \in H, \end{cases} \tag{7.2}$$

where A is self-adjoint and positive, that is, $\langle Ax, x\rangle \geq m\,\|x\|^2$ for some $m \geq 0$ and all $x \in \mathfrak{D}(A)$.

The task is to find *vector-valued* solutions $x : [0,\tau) \to H$ of the equation $\dot{x} + Ax = 0$ satisfying $x \in C^1((0,\tau); H) \cap C([0,\tau); H)$ and the initial value condition $x(0) = x_0$. If A is positive, then any solution $x : [0,\tau) \to H$ of problem (7.2) satisfies the estimate

$$\|x(t)\| \leq \|x(0)\| \quad (t > 0), \tag{7.3}$$

which is a consequence of the inequality $\|x(t)\|^2 = -2\,\langle Ax, x\rangle \leq 0$ and therefore causes the solution to be unique.

To verify the existence of a solution for (7.2), let $(P_\lambda)_{\lambda\in\mathbb{R}}$ be the spectral resolution of the self-adjoint operator A given by the spectral theorem. We notice $P_\lambda = 0$ for $\lambda < 0$, since A is positive. Then the functional calculus applied to the continuous function $t \mapsto e^{-t}$ gives rise to the spectral operator

$$L_t(x) := e^{-tA}x = \int_0^\infty e^{-\lambda t}\,dP_\lambda\,x \qquad (t \geq 0,\ x \in H).$$

We apparently have $L_0 = 1_X$, and (7.3) implies $\|L_t\| \leq 1$ for $t > 0$.

In conclusion, the existence of the solution of (7.2) is ensured by the following

THEOREM 7.4. The function $x : t \mapsto L_t(x_0)$ is continuous in $[0,\infty)$, continuously differentiable in $(0,\infty)$ and forms a solution of (7.2).

2. The *homogeneous Cauchy problem of second order* consists of solving the initial value problem

$$\begin{cases} \ddot{x}(t) + A\,x(t) = 0, \\ x(0) = x_0, \quad \dot{x}(0) = x_1, \end{cases} \tag{7.4}$$

where the operator A again is self-adjoint and positive. We also suppose $x_0 \in \mathfrak{D}(A)$, but the element x_1 may belong to any other linear subspace of X. A function x is called a classical solution of this problem, if $x \in C([0,\tau], H) \cap C^2((0,\tau), H)$ and the initial conditions are satisfied.

The functional calculus for self-adjoint linear operators may also be applied to this problem. Preliminary, each solution x of (7.4) satisfies the *energy conservation law*

$$\|\dot{x}(t)\|^2 + \|\sqrt{A}\, x(t)\|^2 = \|\dot{x}(0)\|^2 + \|\sqrt{A}\, x(0)\|^2 \quad (0 < t < \tau),$$

which implies that the problem owns mostly one solution. Let $\{P_\lambda\}_{\lambda \in \mathbb{R}}$ be the spectral resolution of A. Then we define a function $x : [0,\tau) \to H$ by means of the spectral operator L_t, which itself is induced by the functions $t \mapsto \cos(t)$ and $t \mapsto \sin(t)$.

Indeed, for $0 \leq t < \tau$ the operator $x \mapsto L_t x$ is built up as the sum of two single spectral operators, namely

$$x(t) := L_t(x_0, x_1) := \int_0^\infty \cos\sqrt{\lambda}t\, dP_\lambda\, x_0 + \int_0^\infty \frac{\sin\sqrt{\lambda}t}{\sqrt{\lambda}}\, dP_\lambda\, x_1.$$

Having defined the family of linear operators $\{L_t(x_0, x_1)\}$ this way, we obtain the following existence and uniqueness theorem for the second-order homogeneous Cauchy problem:

THEOREM 7.5. The mapping $t \mapsto L_t(x_0, x_1)$ is continuous in $[0,\tau)$, twice continuously differentiable in $(0,\tau)$ and constitutes the unique solution of problem (7.4).

3. By the *abstract Schrödinger equation* is meant the linear evolution equation

$$\begin{cases} \dot{x}(t) + iA\, x(t) = 0, \\ x(0) = x_0 \in H, \end{cases} \tag{7.5}$$

where A is a self-adjoint linear operator. Each solution x of (7.5) satisfies the *energy equation*

$$\|x(t)\| = \|x(0)\| \tag{7.6}$$

for $t \in \mathbb{R}$, which immediately follows from

$$\frac{d}{dt}\, \|x(t)\|^2 = \langle -iAx(t), x(t)\rangle + \langle x(t), -iAx(t)\rangle = 0.$$

If an initial value $x(0)=x_0$ is given, then (7.6) implies that the solution of the initial-value problem for (7.5) is unique.

To construct a solution by the functional calculus method for self-adjoint operators, we define the spectral operator

$$L_t(x) := e^{-itA}x = \int_{-\infty}^{\infty} e^{-it\lambda} dP_\lambda\, x \qquad (t\in\mathbb{R}, x\in H),$$

where P_λ again is the spectral resolution of A having support in the whole real line. Then the family of operators L_t yields the solution of the abstract Schrödinger problem:

THEOREM 7.6. The mapping $t\in\mathbb{R}\mapsto L_t(x_0)$ is the unique solution of the Cauchy problem for (7.5).

Moreover, for each $t\in\mathbb{R}$ the operator L_t is unitary, and we have $L_t^{-1}=L_{-t}$ for all $t\in\mathbb{R}$.

8. Operators with Pure Point Spectrum

Recall from section 5 that we have introduced various notations of essential spectrum and especially the Browder spectrum $\sigma_{ess,6}(A)$. On the other hand, the *discrete spectrum* $\sigma_d(A)$ of a linear operator A acting in a complex Banach space is defined to be the relative complement of $\sigma_{ess,6}(A)$ with respect to $\sigma(A)$. Hence the set $\sigma_d(A)$ contains those points $\lambda\in\sigma(A)$, which are isolated in $\sigma(A)$ and for which the rank of the *Riesz projection* P_λ is finite. In this case we have a disjoint decomposition of $\sigma(A)$ according to

$$\sigma(A)=\sigma_d(A)\sqcup\sigma_{ess,6}(A).$$

where P_λ is defined by the *Dunford integral*

$$P_\lambda(x) := \frac{1}{2\pi i}\int_{\gamma_\lambda} R(A,\zeta)x\, d\zeta \quad (x\in X) \tag{8.1}$$

and γ_λ is a closed continuous path enclosing λ but no other values in $\sigma(A)$. One may also introduce the set $\sigma_d(A)$ as the collection of isolated points $\lambda\in\sigma(A)$ having finite algebraical multiplicity (this magnitude is the Hamel dimension of the linear space

$$E_{gen}(A,\lambda) := \cup_{n\in\mathbb{N}}E(A^n,\lambda)$$

of *generalized* eigenvalues of A). Elements of the set $\sigma_d(A)$ are called *normal eigenvalues.* It is meaningful to say that A has *pure point spectrum* if $\sigma(A)=\sigma_d(A)$ and $\sigma_{ess,6}(A)=\emptyset$, respectively.

The theory of linear operators having pure point spectrum mainly relies on two fundamental theorems:

1. the *spectral mapping theorem for the discrete spectrum*, quoting that the discrete spectrum of any resolvent $R(A,\zeta)$ depends on that of A according to
$$\sigma_d(R(A,\zeta)) \setminus \{0\} \;=\; \{(\lambda-\zeta)^{-1} : \lambda \in \sigma_d(A)\};$$
2. the *Riesz-Schauder spectral theorem*, which illustrates that the spectral theory of compact linear operators does in principle not differ from that of linear transformations in the finite-dimensional linear spaces $\mathbb{C}^d$. Before quoting this theorem, let us have a short look to this case, which can completely be covered by the theory of square matrices.

   In finite-dimensional linear algebra, the spectral theory begins with the study of eigenvalues and eigenvectors of a square matrix $\mathcal{A} \in \mathbb{C}^{d\times d}$. Remember that an eigenvalue λ of $\mathcal{A}$ is a complex number satisfying $\mathcal{A}x = \lambda x$ for some nonzero vector $x \in \mathbb{C}^d$, called an eigenvector corresponding to λ. The set of all eigenvalues of $\mathcal{A}$, counting multiplicities, forms the spectrum $\sigma(\mathcal{A})$. Moreover, the eigenvalues of $\mathcal{A}$ are precisely the roots of its characteristic polynomial $p_{\mathcal{A}}(\lambda) = det(\lambda - \mathcal{A})$, a monic polynomial of degree d. The algebraic multiplicity of an eigenvalue λ is the multiplicity of λ as a root of $p_{\mathcal{A}}(\lambda)$. The geometric multiplicity of λ, however, is the dimension of the eigenspace $\mathfrak{N}(\mathcal{A}-\lambda)$, which is at most the algebraic multiplicity.

   A matrix $\mathcal{A}$ is diagonalizable if and only if for every eigenvalue geometric multiplicity equals the algebraic multiplicity. When $\mathcal{A}$ is not diagonalizable, its structure is captured by the *Jordan canonical form*, which decomposes $\mathbb{C}^d$ into generalized eigenspaces. For an eigenvalue λ, the generalized eigenspace is nothing but $\mathfrak{N}(\mathcal{A}-\lambda)^m$, where m is the *index* of λ, defined as the smallest positive integer such that $\mathfrak{N}(\mathcal{A}-\lambda)^m = \mathfrak{N}(\mathcal{A}-\lambda)^{m+1}$. This index equals the size of the largest Jordan block associated with λ. The space $\mathbb{C}^n$ then decomposes as a direct sum of these generalized eigenspaces over all distinct eigenvalues. The *Jordan canonical form theorem* states that every square matrix $\mathcal{A}$ over an algebraically closed field such as $\mathbb{C}$ is similar to a unique (up to permutation of blocks) Jordan matrix J, meaning there exists an invertible matrix P such that $\mathcal{A} = PJP^{-1}$. The matrix J is block-diagonal, consisting of Jordan blocks $J_k(\lambda)$ for each eigenvalue λ, where a $k \times k$ Jordan block has the value λ on the main diagonal and the value 1 on the superdiagonal. This form was established by C. JORDAN in his 1870 treatise. A simple example of a non-diagonalizable matrix is

$$\mathcal{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

which has the eigenvalue 0 with algebraic multiplicity 2 but geometric multiplicity 1.

A linear operator k acting from a normed linear space X into a normed linear space Y is called *compact*, if it maps the unit ball $B_1(0) \subset X$ into a pre-compact subset of Y, that is, the closure of the image $k(B_1(0))$ is compact. Then the announced theorem elaborated by F. RIESZ and J. SCHAUDER reads as follows:

THEOREM 8.1. Let X be a complex Banach space and $k : X \to X$ be compact. Then the following assertions hold:
(a) The spectrum $\sigma(k) \subset \mathbb{C}$ is mostly countable.
(b) $\lambda = 0$ is the only possible limit point of $\sigma(k)$.
(c) If X is infinite-dimensional, then $0 \in \sigma(k)$.
(d) Each $0 \neq \lambda \in \sigma(k)$ is an eigenvalue of k with $\alpha(k - \lambda) < \infty$.
(e) Every spectral value $\lambda \neq 0$ is a pole of finite order of the resolvent function $\zeta \in \rho(k) \mapsto R(k, \zeta)$.

Both theorems together imply that a linear operator A has pure point spectrum if and only if at least one of the resolvents is compact, hence A is said to *have compact resolvents* in this case. But notice that if one resolvent of A is compact, then all other resolvents are so, which is a consequence of the resolvent equation (4.1).

Furthermore, a closed linear operator A has compact resolvents if and only if the embedding of the Banach space $(\mathfrak{D}(A), \|\cdot\|_A)$ into the space $(X, \|\cdot\|)$ is compact, where the domain of definition $\mathfrak{D}(A)$ is furnished with the *operator norm* $\|x\|_A := \|x\| + \|Ax\|$.

The eigenvalues of a linear operator A with pure point spectrum are mostly countable and can be ordered by their moduli, that is, there is a sequence $\{\lambda_n\}_{n \in \mathbb{N}}$ consisting of eigenvalues of A such that

$$|\lambda_1| \leq |\lambda_2| \leq |\lambda_3| \leq \dots .$$

But more can be said in the case, where $X = H$ is a Hilbert space and the operator A is symmetric. Then all eigenvalues λ_n are real-valued and consequently A proves to be self-adjoint. If in addition A is half-bounded with lower bound m, we may improve the order of the sequence of eigenvalues to

$$m \leq \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq \dots .$$

To the end, if the underlying Hilbert space is separable, then the collection of eigenvectors $\{x_n\}_{n \in \mathbb{N}}$ of the operator A forms an orthonormal base. This statement is the content of the celebrated *Hilbert-Schmidt spectral theorem*:

THEOREM 8.2. If the linear operator k acting in a separable Hibert space H is compact and symmetric, then the collection $\mathfrak{O} \subset H$ of normed eigenvectors of k is complete and constitutes an orthonormal system in H, respectively.

We conclude from this theorem, often also denoted the *discrete spectral theorem*, that a symmetric linear operator A with pure point spectrum admits the *spectral decomposition*

$$A\,x = \sum_{n=1}^{\infty} \lambda_n \langle x, x_n \rangle \, x_n$$

for all $x \in \mathfrak{D}(A)$, hence the domain of definition of A can be described according to

$$\mathfrak{D}(A) = \{x \in H : \sum_{n=1}^{\infty} \lambda_n^2 \, |\langle x, x_n \rangle|^2 < \infty\}.$$

The discrete spectral theorem can be seen as a special case of the general spectral theorem 7.2 for unbounded self-adjoint operators. Indeed, the spectral resolution of a self-adjoint operator with pure point spectrum consists of the projections

$$P_\lambda := \sum_{\lambda_k < \lambda} \pi_{\lambda_k} \qquad (\lambda \in \mathbb{R}),$$

where π_{λ_k} is the projection onto the eigenspace E_{λ_k} of the eigenvalue λ_k, hence the mappings P_λ are constant for $\lambda \in [\lambda_k, \lambda_{k+1})$.

Let us in the second part of this section turn back to the class of Fredholm operators and illustrate the interplay of Fredholm theory with the theory of linear operators having pure point spectrum.

If $k: X \to X$ is compact, then it follows from theorem 8.1 (d), that the operator 1_X+k has finite nullity index and closed range, hence it is upper semi-Fredholm. But this is also true for the operator $1_{X^*}+k^*$, since k^* is compact, too, and both properties together imply that 1_X+k is Fredholm with index $\chi(1_X+k)=0$.

The following computation is of central meaning and for example useful in solving the *Dirichlet problem* for linear-elliptic differential operators. Consider the linear equation $Ax=y$, where X is a Banach space and $A: \mathfrak{D}(A) \subset X \to X$ is a linear operator with pure point spectrum. Then for $\zeta \in \rho(A)$ we surely have

$$A\,x \;=\; (A-\zeta)\,x + \zeta\,x \;=\; y,$$

and applying the mapping $(A-\zeta)^{-1}$ from the left yields

$$x \;+\; \zeta(A-\zeta)^{-1}\,x \;=\; (A-\zeta)^{-1}\,y \;=:\; \tilde{y}$$

or equivalently $x+\tilde{k}\,x=\tilde{y}$, where $\tilde{k} := \zeta(A-\zeta)^{-1}$ is compact. The equations $A\,x=y$ and $(1_X+\tilde{k})\,x=\tilde{y}$ are called *equivalent to each other* in the sense that the element x is a solution of the first one if and only if it is a solution of the second one.

We summarize the result of the previous computation to the following

THEOREM 8.3. Let $A: \mathfrak{D}(A) \subset X \to X$ be an operator with pure point spectrum. Then the equation $Ax=y$ is equivalent to an equation of the form $(1_X+\tilde{k})x=\tilde{y}$, where the operator $\tilde{k}$ is compact.

Since $1_X+\tilde{k}$ is Fredholm with vanishing index, FREDHOLM's alternative theorem 5.6 is applicable and therefore provides also assertions on the solution of the equation $Ax=y$.

Let us finalize the elaborations on linear equations with a complete collection of concepts regarding the solvability of $A\,x=y$, where A is assumed to be closed and acting between some Banach spaces X and Y. The equation $A\,x=y$ is called

(a) *densely solvable* ([10]), if $\overline{\mathfrak{R}(A)}=Y$;
(b) *normally solvable* ([10]), if $\overline{\mathfrak{R}(A)}=\mathfrak{R}(A)$;
(c) *correctly solvable* ([10]), if A is bounded below;
(d) *semi-solvable* ([2]), if A is Fredholm;
(e) *almost solvable* or *regularly solvable* ([2]), if A is semi-solvable and $\chi(A)=0$;

(f) *uniquely solvable* ([10]), if A is injective;
(g) *everywhere solvable* ([10]), if A is surjective;
(h) *solvable* ([2]), if it is uniquely and everywhere solvable;
(i) *well-posed*, if it is solvable and A^{-1} is continuous, or equivalently, if $0 \in \rho(A)$;
(j) *completely solvable* ([2]) or *compactly solvable* if it is solvable and A^{-1} is compact.

9. Infinitesimal Generators of Semigroups

This section contains a short outlook to *semigroup theory*, the subject that deals with families of bounded linear operators fulfilling two fundamental features. The theory of semigroups proves to be one of the most suitable tools in the study of linear evolution equations.

Given a complex Banach space X, a *strongly continuous semigroup of bounded linear operators* (shortly denoted a C_0*-semigroup*) is an operator-valued mapping $S:[0,\infty) \to \mathscr{L}(X)$, also denoted $\{S(t)\}_{t\geq 0}$, which distinguishes by the conditions

(G-0) $S(0) = 1_X$;
(G-1) $S(t+s) = S(t) \circ S(s)$ for $t, s \geq 0$ (*semigroup property*);
(G-2) $\lim_{t\downarrow 0} S(t)\, x = x$ for all $x \in X$.

It follows from (G-2), which is nothing but strong continuity from the right in $t=0$, that $S(0)$ is the identity map 1_X. The conditions (G-1) and (G-2) together imply strong continuity from the right of the semigroup even in every $t>0$. In case of $X = \mathbb{R}$, the family $\{\mathcal{E}(t)\}_{t\geq 0}$ of exponential functions $\mathcal{E}(t): x \in \mathbb{R} \mapsto e^{\lambda t x}$ fulfills both conditions, so it serves as an elementary example of a C_0-semigroup.

The totality of strongly continuous semigroups may be classified as follows: for each such semigroup $\{S(t)\}_{t\geq 0}$ there are constants $M \geq 0$ and $\omega \in \mathbb{R}$ such that

$$\|S(t)\| \leq Me^{\omega t} \quad \text{for } t > 0. \tag{9.1}$$

We denote ω the *growth bound* of the semigroup. In the special case, where $(M,\omega) = (0,1)$ and $\|S(t)\| \leq 1$ for all $t \geq 0$, respectively, the semigroup is said to be *contractive*. If $\omega = 0$, however, the semigroup is called *uniformly bounded*.

By the *orbit map* or *trajectory* η_x of a C_0-semigroup $\{S(t)\}_{t\geq 0}$ and associated with the initial value $x \in X$ we mean the vector-valued mapping $\eta_x : [0,\infty) \to X$ defined by $\eta_x(t) := S(t)\, x$. The conditions (G-1) and (G-2) together imply continuity of the map η_x. But η_x is even differentiable under a mild assumption: if $\{S(t)\}_{t\geq 0}$ is strongly continuous, then for each $x \in X$ the mapping η_x is differentiable in $(0, \infty)$ iff η_x is right-differentiable in $t=0$.

The last statement serves as motivation for the following notion: if $\{S(t)\}_{t\geq 0}$ is strongly continuous, its *infinitesimal generator* of the semigroup (briefly denoted the *generator* or the *generating operator*) is defined as the mapping $A : \mathfrak{D}(A) \subset X \to X$ given by the graph

$$\Gamma_A := \{ (x, y) \in X \times X : \; y = \lim_{h \downarrow 0} \frac{1}{h}(S(h)x - x) \text{ exists} \}.$$

Hence the domain of definition $\mathfrak{D}(A)$ consists of those elements $x \in X$, for which the orbit map η_x is differentiable from the right in $t=0$. But this means that η_x is also (left and right) differentiable in every $t > 0$, which is a consequence of property (G-1).

It is easy to show that the infinitesimal generator of a strongly continuous semigroup is linear, so it forms a linear operator in X.

Our first theorem in this section describes two basic properties of infinitesimal generators:

THEOREM 9.1. The infinitesimal generator of a C_0-semigroup is densely defined and closed.

Hence the generators of strongly continuous semigroups fit into our base class of linear operators considered in section 2.

Functional analysis provides a rich theory on *homogeneous* linear evolution equations of the form

$$\begin{cases} \dot{x}(t) + A\,x(t) = 0, \\ x(0) = x_0 \in X, \end{cases} \tag{9.2}$$

where A is a densely defined and closed linear operator acting in a Banach space X and with domain of definition $\mathfrak{D}(A) \subset X$.

Like in section 7, it is to find a solution $x : [0, \tau) \to X$ of the equation $\dot{x} + Ax = 0$ satisfying the initial value condition $x(0) = x_0$. This problem has already been considered in the Hilbert space setting, where A was assumed to be self-adjoint and positive, and the solution of the problem (9.2) was constructed by means of the spectral resolution of A and the functional calculus for unbounded self-adjoint operators.

We first have to define concepts of solution for problem (9.2):

1. A vector-valued function $x \in C([0,\tau), \mathfrak{D}(A)) \cap C^1((0,\tau), X)$ is called a *classical solution* of (9.2), if it satisfies the initial condition $x(0) = x_0$ and the differential equation in each point $t \in (0,\tau)$.
2. A continuous function $x : [0,\tau) \to X$ is said to be a *mild solution* of (9.2) if

$$(1)\ \tilde{x}(t) := \int_0^t x(s)\,ds \in \mathfrak{D}(A) \quad \text{and} \quad (2)\ A\,\tilde{x}(t) = x(t) - x_0$$

holds for all $0 < t < \tau$, where the integral on the right-hand side of (1) is meant in the Riemannian sense. One is led to the notion of mild solution in a natural way by integrating both sides of the equation $\dot{x} + A\,x = 0$, which then becomes into

$$\int_0^t \dot{x} ds \;=\; x(t) - x(0) \;=\; -\int_0^t A\,x(s)\,ds.$$

If the operator A is closed, then the action of A and the execution of the integral commutes for continuous functions $x : [0,t] \to \mathfrak{D}(A)$, that is,

$$\int_0^t A\,x(s)\,ds \;=\; A\int_0^t x(s)\,ds,$$

and to the end we obtain the equation

$$x(t) \;=\; x(0) - A\int_0^t x(s)\,ds \qquad (t \in (0,\tau)).$$

Notice that each classical solution is a mild solution, and reversely a mild solution is classical iff it is continuously differentiable in $(0,\tau)$. Thus classical and mild solutions of homogeneous Cauchy problems differ with respect to this regularity condition. It should also be mentioned that if a solution x is classical, then $x(0) = x_0$ must belong to the domain of definition $\mathfrak{D}(A)$, whereas for a mild solution x the initial value x_0 may be an arbitrary element in X.

The homogeneous Cauchy problem (9.2) is called *correctly posed*, if there is a unique classical solution x for any initial value $x_0 \in \mathfrak{D}(A)$, that is, $x \in C^1((0,\tau); X) \cap C([0,\tau); X)$.

Moreover, the problem is called *well-posed*, if it is correctly posed and the solution x depends continuously on the initial value x_0. In other words, the problem is *well-posed*, if a classical solution exists and for each $\tau > 0$ there is $c_\tau > 0$ such that $\|x(t)\| \le c_\tau\,\|x_0\|$ for all $t \in (0,\tau)$ and all $x_0 \in X$.

The following fundamental theorem describes well-posedness of (9.2) in terms of the property of the linear operator $-A$ to be an infinitesimal generator:

THEOREM 9.2. Let A be a densely defined and closed linear operator acting in a Banach space X. Then the problem (9.2) is well-posed if and only if $-A$ is the infinitesimal generator of a C_0-semigroup.

Indeed, the family $\{S(t)\}_{t\geq 0}$ of linear operators having $-A$ as its infinitesimal generator yields the classical solutions x with initial values $x_0 \in \mathfrak{D}(A)$ of (9.2), that is, $x(t,x_0) = S(t)\,x_0$ for $0 \leq t < \tau$. The proof of this fundamental well-posedness theorem can be found in [5], p. 112 as that of theorem II.6.6.

Let us finally mention that problem (9.2) governed by a densely defined and closed linear operator A is well-posed iff it is correctly posed and $\rho(A) \neq \emptyset$ holds.

Besides the fact that the operator A has to be densely defined and closed, the further conditions on $-A$ to be the infinitesimal generator of a C_0-semigroup can be expressed *(i)* by the location of the resolvent set $\rho(A)$ within the complex plane and *(ii)* by a collection of estimates for the resolvents $R(A,\zeta)$ in a subset of $\rho(A)$.

Let the semigroup $\{S(t)\}_{t\geq 0}$ be strongly continuous. Then for fixed $x \in X$ and $\zeta \in \mathbb{C}$, the value $L_n(\zeta,x)$ defined by

$$L_n(\zeta,x) := \frac{1}{(n-1)!}\int_0^\infty t^{n-1}e^{\zeta t}S(t)x\,dt \tag{9.3}$$

is called the (right side) *Laplace transform* of order n of the semigroup with respect to x and ζ. This expression has to be understood as improper vector-valued Riemann-Stieltjes integral.

We cannot expect that $L_n(\zeta,x)$ exists without assuming further conditions. But $L_n(\zeta,x)$ is surely well-defined for values ζ with $\operatorname{Re}\zeta < \omega$ and all $x \in X$. Hence the mapping $L_n(\zeta)x := L_n(\zeta,x)$ constitutes a linear operator in X with norm estimate

$$\|L_n(\zeta,\cdot)\| \leq \frac{1}{(n-1)!}\int_0^\infty t^{n-1}e^{-\operatorname{Re}\zeta t}e^{\omega t}\,dt = \frac{M}{(\operatorname{Re}\zeta-\omega)^n}.$$

A short computation shows that

$$\begin{cases} L_1(\zeta)(\zeta-A) = (\zeta-A)L_1(\zeta) = 1_X, \\ L_n(\zeta)(\zeta-A) = (\zeta-A)L_n(\zeta) = L_{n-1}(\zeta), \end{cases}$$

which implies that $(-\infty,\omega) \subset \rho(A)$ and $R(A,\zeta) = L_n(\zeta)$ holds for all $n \in \mathbb{N}$.

Reversely, the question arises, under which conditions a linear operator $-A$ acting in X forms the infinitesimal generator of any C_0-semigroup. For bounded operators the answer can be given quickly: if $\|A\|$ is the operator norm of A, then the mappings

$$\exp(-A)(t)\,x := \sum_{n=0}^{\infty} \frac{(-tA)^n}{n!}\,x$$

for $x \in X$ and $t > 0$ form a strongly continuous semigroup.

The general answer, however, is given by the *Feller-Miyadera-Hille-Yosida theorem*, which characterizes properties of the operator $-A$ to be the infinitesimal generator of a C_0-semigroup:

THEOREM 9.3. Let A be a densely defined and closed linear operator acting in an arbitrary Banach space X. Then the following assertions are equivalent to each other:

(a) The operator $-A$ generates a strongly continuous semigroup of type (ω, M).

(b) The resolvent set of A fulfills $(-\infty,\omega) \subset \rho(A)$ and the resolvents $R(A,\zeta)$ satisfy the *Hille-Yosida estimates*

$$\|R(A,\zeta)^n\| \le \frac{M}{(\omega-\zeta)^n} \qquad (n \in \mathbb{N},\ \zeta < \omega). \tag{9.4}$$

The theorem shows that the already known conditions on $-A$ to be a generating operator are also sufficient.

However, since the number of conditions on the operator A is infinite, this theorem has limited worth with respect to applications and even in the simplest cases. But the original theorem for generators of *contraction semigroups* ($\omega = 0$ and $M = 1$) due to HILLE and YOSIDA quotes that in this case only the first inequality ($n = 1$) in (9.4) has to be verified:

THEOREM 9.4. The operator $-A$ acting in a Banach space X is the infinitesimal generator of a C_0-semigroup of *contractions* iff

(a) A is densely defined and closed,

(b) the interval $(-\infty, 0)$ belongs to the resolvent set $\rho(A)$,

(c) the resolvent estimation in the contractional case holds:

$$\|\zeta\, R(A,\zeta)\| \le 1 \quad (\zeta < 0). \tag{9.5}$$

We want to sketch how the assertions (b) and (c) together imply that A generates a contractional C_0-semigroup. First of all, it can be verified that (9.5) implies $\lim_{\zeta\to-\infty}\zeta R(A,\zeta)=x$ for all $x\in\mathfrak{D}(A)$. As a consequence, the so-called *Yosida operators*

$$A_n := nA(n-A)^{-1} = n^2-(A-n)^{-1}-n$$

approximate the operator A as $n\to-\infty$ in the sense that

$$\lim_{n\to-\infty} A_n x = Ax$$

for all $x\in\mathfrak{D}(A)$. Since the operators A_n are bounded, one may define semigroups for each of them by letting

$$e^{-tA_n}x := \sum_{k=0}^{\infty}\frac{(-tA_n)^k}{k!}\,x \qquad (x\in X).$$

It follows that $\|e^{-tA_\zeta}\|\le 1$ for $t>0$, so we obtain even semigroups of contractions. Now the main part of the proof is to show that this sequence of semigroups is Cauchy and the semigroup generated by the operator $-A$ is given by

$$S(t)x := \lim_{n\to-\infty} e^{-tA_n}x \qquad (x\in X). \tag{9.6}$$

To be precise, one first has to verify property (S-1) and (S-2) and second the operator $-A$ to be the generator of the semigroup (9.6).

10. Accretive Operators

The shapes and locations of the spectrum and the numerical range of a densely defined and closed linear operator acting in a complex Hilbert space give rise to introduce a couple of subclasses of such operators, and one of them is the class of *accretive operators.*

A linear operator A acting in H is called *accretive*, if the numerical range $\Theta(A)$ fulfills

$$\Theta(A)\subset\mathbb{C}^+ := \{\zeta\in\mathbb{C}:\operatorname{Re}\zeta\ge 0\},$$

or equivalently, if $\operatorname{Re}\langle Ax,x\rangle\ge 0$ for all $x\in\mathfrak{D}(A)$. Furthermore, the operator A is said to be *quasi-accretive*, if there is $\lambda_0\in\mathbb{R}$ such that $A+\lambda_0$ is accretive.

One verifies that the condition for accretivity is equivalent to

$$\|(A-\zeta)x\| \ge \|\zeta x\| \tag{10.1}$$

for $\zeta<0$ and $x\in\mathfrak{D}(A)$ and also equivalent to

$$\|(A-\operatorname{Re}\zeta)x\| \;\geq\; \|\operatorname{Re}\zeta\, x\|$$

for $\zeta < 0$ and $x \in \mathfrak{D}(A)$. But this means that the operators $A-\zeta$ are bounded from below, that is, they are injective and own closed ranges in case of A being closed.

Finally, the operator A is said to be *m-accretive*, if it is accretive and does not have any proper accretive extension. Equivalently, an accretive operator A is m-accretive, if there is $\zeta_0 \in (-\infty, 0)$ such that $\mathfrak{R}(A-\zeta_0) = H$. In this case the operator $A-\zeta_0$ is actually bijective. Hence we have the estimate

$$\|\, \zeta_0\, R(A, \zeta_0)\, \| \;\leq\; 1,$$

where $R(A, \zeta_0)$ again denotes the resolvent for $\zeta_0 \in \rho(A)$.

Like for self-adjoint operators, the necessarity of the existence of a number $\zeta_0 \in \mathbb{C}$ such that $A-\zeta$ be surjective, is called the *range condition* for m-dissipativity. It follows from the stability theory of semi-Fredholm operators that if such a number ζ_0 exists, then $A-\zeta$ is surjective even for every $\zeta \in (-\infty, 0)$, so ζ_0 may be replaced by any other number $\zeta < 0$.

A m-accretive operator fits into the main class of densely defined and closed linear operators, as the following theorem shows:

THEOREM 10.1. Each m-accretive operator A is densely defined and closed, and the Hilbert adjoint A^* is m-accretive, too.

Some further assertions on m-accretive operators are collected to the following theorem:

THEOREM 10.2. The following statements are equivalent:

(a) A is m-accretive;

(b) $A + i\delta$ is m-accretive for all $\delta \in \mathbb{R}$;

(c) $A + \epsilon$ is m-accretive for all $\epsilon > 0$;

(d) $-\zeta \in \rho(A)$ and $\|(A-\zeta)(A+\zeta)^{-1}\| \leq 1$ for $\zeta > 0$;

(e) $\mathbb{C}^- := \{z \in \mathbb{C} : \operatorname{Re} z < 0\} \subset \rho(A)$ and

$$\|R(A,\zeta)\| \leq \frac{1}{|\operatorname{Re}\zeta|} \qquad (\operatorname{Re}\zeta < 0);$$

(f) A is densely defined and closed, and A^* is m-accretive (this is the inverse statement of theorem 10.1).

The usage of m-accretivity is illustrated by the next theorem. By means of this concept, the celebrated *Lumer-Phillips theorem* classifies those linear operators, which are infinitesimal generators of contractive C_0-semigroups:

THEOREM 10.3. Let A be a linear operator in Hilbert space. Then $-A$ generates a contractive C_0-semigroup iff A is m-accretive.

The proof of this theorem relies on the classical Hille-Yosida theorem for C_0-semigroups of type $(0,1)$. Indeed, A is m-accretive, then A is densely defined, closed and $-A$ fullfills the estimate (10.1), hence $-A$ is the generator of a contractive C_0-semigroup.

Reversely, if $-A$ is the infinitesimal generator of a C_0-semigroup of contractions, then we have $(0,\infty)\subset\rho(A)$ and therefore $\mathfrak{R}(A-\zeta)=H$ for all $\zeta>0$. In addition, if $x\in\mathfrak{D}(A)$, then

$$|\langle S(t)x,x\rangle| \;\leq\; \|S(t)x\|\,\|x\| \;\leq\; \|x\|^2$$

and

$$\mathrm{Re}\langle S(t)x-x,x\rangle \;=\; \mathrm{Re}\langle S(t)x,x\rangle \;-\; \|x\|^2 \;\leq\; 0.$$

Dividing this line by $t>0$ and letting $t\downarrow 0$ yields $\mathrm{Re}\langle Ax,x\rangle\leq 0$, which shows that $-A$ is accretive, hence in summary the operator A even proves to be m-accretive.

Similar to the first-order Cauchy problem, a function x is called a classical solution of the *second-order* homogeneous Cauchy problem

$$\left\{\begin{array}{rcl} \ddot{x}(t)+Ax(t) &=& 0,\\ x(0) &=& x_0\in X,\\ \dot{x}(0) &=& x_1\in X, \end{array}\right. \tag{10.2}$$

if $x\in C([0,\tau),\mathfrak{D}(A))\cap C^2((0,\tau),X)$ and satisfies both initial conditions as well as the differential equation in each point $t\in(0,\tau)$.

To apply results from semigroup theory to the second order Cauchy problem, let $U:=(x,\dot{x})$. We suppose that A is a *positive* and *self-adjoint* linear operator in a Hilbert space H. Then the differential equation $\ddot{x}(t)+A\,x(t)=0$ in (10.2) can be written as first order problem

$$\left\{\begin{array}{l} U'(t) \;=\; B\,U(t)\\ U(0)=(x_0,x_1), \end{array}\right. \tag{10.3}$$

where the operator B is defined by

$$B := \begin{pmatrix} 0 & 1 \\ \sqrt{A} & 0 \end{pmatrix}. \tag{10.4}$$

The main idea in proving existence and uniqueness of a solution is to show that the operator B defined on a suitable chosen domain $\mathfrak{D}(B)$ is the infinitesimal generator of a semigroup of contractions. Its domain of definition $\mathfrak{D}(B)$ is given by

$$\mathfrak{D}(B) := \mathfrak{D}(A) \times \mathfrak{D}(\sqrt{A}) \subset \mathfrak{D}(\sqrt{A})_{\|\cdot\|_A} \times H,$$

where $\mathfrak{D}(\sqrt{A})_{\|\cdot\|_A}$ is the Hilbert space $\mathfrak{D}(\sqrt{A})$ furnished with the graph norm.

The following theorem is for example proved in [7], p. 111:

THEOREM 10.4. Let the linear operator A be positive self-adjoint and the operator B be defined by (10.4). Then B is m-accretive, hence the Cauchy problem (10.3) is well-posed.

In the Hilbert space setting an accretive operator often appears as the form operator associated with a *quasi-coercive* and continuous sesquilinear form $\mathfrak{a}\ V \times V \to \mathbb{C}$, defined by means of a Hilbert space triple (V, H, V_a^*).

We want to make this assertion precise. Each Hilbert pair (V, H) with continuous embedding $\iota\colon V \to H$ gives rise to introduce the *transposed pair* (H_a^*, V_a^*), where the conjugate linear space H_a^* is continuously and densely embedded into the conjugate linear space V_a^* by a further continuous embedding $\iota_2\colon H_a^* \to V_a^*$. Then, by identification of the spaces H and H_a^* via the Riesz map, combining both Hilbert pairs leads to the triple

$$V \overset{\iota_1}{\hookrightarrow} H \overset{\iota_2}{\hookrightarrow} V_a^*, \tag{10.5}$$

sometimes called a *Gelfand triple*, an *evolution triple* or a *rigged Hilbert space*. The space H is often denoted the *pivot space* of (10.5).

Gelfand triples own a remarkable property. Let $(\cdot\,,\cdot)$ and $\langle\cdot\,,\cdot\rangle$ be the inner products in V and H, respectively, then for fixed chosen $x \in H$ the action of the antilinear linear functional $f_x := \iota_2\, x \in V_a^*$ to any element $v \in V$ is represented by the inner product of x and v in H, i. e. $f_x(v) = \langle x, \iota_1\, v\rangle$ holds for all $v \in V$.

Furthermore, a continuous sesquilinear form $\mathfrak{a}: V \times V \to \mathbb{C}$ is called *coercive* if there is $\theta > 0$ such that

$$\operatorname{Re}\, \mathfrak{a}(x,x) \geq \theta\, \|x\|^2 \qquad \text{for all } x \in V.$$

The celebrated *Lax-Milgram theorem* claims, that if $\mathfrak{a}$ is coercive, then the *variational equation*

$$\mathfrak{a}(x,y) = f(y) \qquad (y \in V)$$

owns for each $f \in V_a^*$ exactly one solution $x \in V$. In other words, the representation operator $\mathcal{A}_{\mathfrak{a}}: V \to V_a^*$, defined by

$$(\mathcal{A}_{\mathfrak{a}} x)(y) := \mathfrak{a}(x,y) \qquad (x, y \in V)$$

is bijective. Moreover, the mapping $\mathcal{A}_{\mathfrak{a}}$ has a bounded inverse with norm $\|\mathcal{A}_{\mathfrak{a}}^{-1}\| \leq \theta^{-1}$, hence the equation $A_{\mathfrak{a}} x = f$ is well-posed.

In applications it is often the case that sesquilinear forms are merely continuous but not coercive. However, if a Hilbert pair (V, H) is given, it is possible to make a form $\mathfrak{a}$ coercive by adding an additional term, which is just the multiplicity of the inner product in H with a real number. More precisely, if $\mathfrak{a}: V \times V \to \mathbb{C}$ is continuous and the slightly modified form

$$\mathfrak{a}_\lambda(u,v) := \mathfrak{a}(u,v) + \lambda\langle \iota_1 u, \iota_1 v\rangle \quad (u, v \in V)$$

is coercive for some $\lambda \in \mathbb{R}$, that is, there is $\theta > 0$ such that

$$\operatorname{Re}\, \mathfrak{a}(u,u) + \lambda\, \|\iota_1 u\|_H^2 \geq \theta\, \|u\|_V^2 \quad (u \in V),$$

then $\mathfrak{a}$ is called *quasi-coercive*, *H-elliptic* or simply a *Gårding form*. It should be clear that if $\mathfrak{a}_{\lambda_0}$ is coercive for some $\lambda_0 \in \mathbb{R}$, then $\mathfrak{a}_\lambda$ is so for $\lambda > \lambda_0$.

Quasi-coercive sesquilinear forms can be used to define densely defined and closed linear operators acting in the pivot space H of a Gelfand triple. To elaborate this assertion, let $\mathfrak{a}$ be a sesquilinear form defined on a Hilbert space V, which is the left-hand side of a Gelfand triple (V, H, V_a^*) with continuous embeddings $\iota_1: V \to H$ and $\iota_2: H \to V_a^*$. It is our aim to introduce a linear operator $A_{\mathfrak{a}}$ with domain space H and range space H. This will be done by restricting the form operator $\mathcal{A}_{\mathfrak{a}}: V \to V_a^*$ associated with $\mathfrak{a}$ to those elements $u \in V$, for which $\mathcal{A}_{\mathfrak{a}}\, u \in H$. Indeed, the domain of definition $\mathfrak{D}(A_{\mathfrak{a}})$ is defined to be the subspace of those elements $u \in V$, for which $\mathcal{A}_{\mathfrak{a}}\, u \in \iota_2(H) \subset V_a^*$ holds, briefly

$$\mathfrak{D}(A_{\mathfrak{a}}) := \{u \in V : \mathcal{A}_{\mathfrak{a}} u \in H\}.$$

The set $\mathfrak{D}(A_{\mathfrak{a}})$ can be regarded as a linear subspace of H by means of the embedding ι_1. Then for $u \in \mathfrak{D}(A_{\mathfrak{a}})$ there is $y \in H$ such that $\iota_2 y = \mathcal{A}_{\mathfrak{a}} u$. We set $x := \iota_1 u$ and $A_{\mathfrak{a}} x := y$ for $x \in \mathfrak{D}(A_{\mathfrak{a}})$ and denote $A_{\mathfrak{a}}$ the *operator in H associated with* $(V, \mathfrak{a})$, or shortly, the *variational operator* derived from $\mathfrak{a}$. Another phrase is to say that $A_{\mathfrak{a}}$ is the *part of $\mathcal{A}_{\mathfrak{a}}$ in H*. Clearly, we have $\mathfrak{a}x, y = \langle Ax, y\rangle$ for all $x \in \mathfrak{D}(A)$ and $y \in V$.

Our last theorem in this section collects a couple of properties of the operator $A_{\mathfrak{a}}$ associated with a quasi-coercive form $\mathfrak{a}$:

THEOREM 10.5. Let (V, H, V_a^*) be a triple of Hilbert spaces and the sesquilinear form $\mathfrak{a}$ be continuous and quasi-coercive on V with parameter $\lambda_0 \in \mathbb{R}$. Let $A_{\mathfrak{a}}$ be the operator associated with $\mathfrak{a}$. Then the following assertions are true:

(a) $A_{\mathfrak{a}}$ is closed.
(b) $A_{\mathfrak{a}}$ is densely defined in H.
(c) If the embedding of V into H is *compact*, then the operator $A_{\mathfrak{a}} + \lambda$ owns a compact inverse $(A_{\mathfrak{a}} + \lambda)^{-1}$ for each $\operatorname{Re}\lambda \geq \lambda_0$.
(d) The operator $A_{\mathfrak{a}} + \lambda$ is m-accretive for each $\operatorname{Re}\lambda \geq \lambda_0$.

It is usual to call a linear operator *regularly accretive*, if it is the operator associated with a quasi-coercive sesquilinear form.

11. Sectorial Operators and Analytic Semigroups

The concept of sectorialness is helpful to study semigroups, which are *analytic* in some sense and form the solution of so-called *parabolic* evolution equations, whose solutions are infinitely differentiable in every $t > 0$.

A linear operator A acting in a complex Banach space X is called *sectorial*, if there is $\psi \in [0, \pi)$ and $\omega \in \mathbb{R}$ such that

(S-1) the spectrum $\sigma(A)$ is contained in the closure $\overline{\Sigma_{\psi,\omega}}$ of a right-open *sector*

$$\Sigma_{\psi,\omega} := \{\zeta \in \mathbb{C} : \zeta \neq \omega, arg(\zeta - \omega) < \psi\}$$

with vertex ω and semi-angle ψ; this implies that $\rho(A)$, the resolvent set of A, contains the complementary and left-open sector $\Lambda_{\pi-\psi,\omega} := \mathbb{C} \setminus \overline{\Sigma_{\psi,\omega}}$;

(S-2) given the angle $\phi < \pi$, for every $\psi < \phi$ the following resolvent estimation holds:

$$\sup \{ \| \zeta R(A,\zeta) \| : \zeta \in \Lambda_{\pi-\psi,\omega} \} < \infty ; \qquad (11.1)$$

Moreover, the operator A is called *sectorial*, if (S-1) and (S-2) hold for some pair $(\psi,\omega) \in [0,\pi) \times \mathbb{R}$. Notice that if A is sectorial with respect to the angle ψ, then there is $0 < \epsilon < \psi$ such that A is also sectorial with respect to the smaller angle $\psi - \epsilon$. The infimum ψ_0 of all these angles is called the *angle of sectoriality* of A.

It should be clear that each sectorial operator is quasi-m-accretive if the resolvent estimate (11.1) holds with $M \leq 1$.

The following theorem is useful to ensure sectorialness:

THEOREM 11.1. Let A be a regular linear operator in X with domain of definition $\mathfrak{D}(A) \subset X$ having the following properties:

(a) $\rho(A)$ contains $\mathbb{C}_\omega := \{\zeta \in \mathbb{C} : \operatorname{Re}\zeta \leq \omega\}$ for some $\omega \in \mathbb{R}$;

(b) there is $M > 0$ such that $\|\zeta\, R(A,\zeta)\| \leq M$ for all $\zeta \in \mathbb{C}_\omega$.

Then A is sectorial with vertex ω.

The proof of this theorem can be found in [11], p. 43.

We next quote two properties of a sectorial linear operator A, when acting in a Hilbert space H:

1. A is densely defined and closed.
2. $H = \mathfrak{N}(A) \oplus \overline{\mathfrak{R}(A)}$.

The second property also holds if A is acting in a reflexive Banach space.

One of the goals in this section is to sketch a functional calculus for a sectorial linear operator A, which especially enables us to derive from A a family of bounded linear operators $\{e^{-tA}\}_{t\geq 0}$.

More precisely, if A is sectorial, then the mapping $t \mapsto e^{-tA}$ is defined by means of

$$e^{-tA}x := \frac{1}{2\pi i} \int_\gamma e^{-t\zeta}\, R(A,\zeta)x\, d\zeta \qquad (x \in H), \qquad (11.2)$$

which generalizes the *Riesz-Dunford formula* (4.6). Here by the symbol γ is meant a rectifiable path in the complex plane that drifts within the sector $S_{\phi,\omega}$ and includes the spectrum of A.

In fact, the family of mappings $\{e^{-tA}\}_{t\geq 0}$ given by (11.2) constitutes a semigroup of bounded linear operators, which is strongly continuous if A is assumed to be densely defined.

Recall that a semigroup of bounded linear operators is usually parametrized by the half line $[0,\infty)$. But it is also customary to study possible extensions of a semigroup to a symmetric sector

$$\Sigma_\theta := \{z = re^{i\alpha} \in \mathbb{C} : r > 0\,, |\alpha| < \theta\}$$

of angle 2θ for some $\theta \in (0, \frac{\pi}{2}]$. A semigroup $\{S(t)\}_{t\geq 0}$ is called *analytic* if

1. there is $\theta \in (0, \frac{\pi}{2})$ and an analytic extension $S' : \Sigma_\theta \to \mathscr{L}(X)$ of $\{S(t)\}_{t\geq 0}$, which is *locally bounded*, that is,

$$\sup\{\, \|S'(z)\| : z \in \Sigma_\theta,\ \|z\| \leq 1\} < \infty,$$

2. the semigroup property $S(t+s) = S(t) \circ S(s)$ holds for $s, t \in \Sigma_\theta$.

The advantage of working with this class of semigroups instead of the strongly continuous one is given by the following features of an analytic semigroup.

1. $Ax \in \mathfrak{D}(A)$ for each $x \in \mathfrak{D}(A)$.
2. The orbits η_x $(x \in X)$ of an analytic semigroup are indefinitely differentiable.

Hence, if A is the generator of an analytic semigroup, then the solutions x of the the differential equation $\dot{x} + Ax = y$ are smooth in all intervals $(0,\tau)$, and by this reason the differential equation is called *parabolic.*

Remember that the functional calculus for a sectorial operator A enables to define the semigroup $\{e^{-tA}\}_{t\geq 0}$ by a contour integral around the spectrum $\sigma(A)$. But more is true in this case:

THEOREM 11.2. A is sectorial, iff $\{e^{-tA}\}_{t\geq 0}$ is analytic.

Let us now consider the case, where the analytic semigroup $\{S(t)\}_{t\geq 0}$ is contractive in a sector Σ_θ, that is, $\|S(\zeta)\| \leq 1$ holds for all $\zeta \in \Sigma_\theta$, and the semigroup is called *sectorially contractive analytic* of angle θ,

The generation theorem for the class of sectorially contractive analytic semigroups reads as follows:

THEOREM 11.3. Let A be densely defined and let $\theta \in (0, \frac{\pi}{2}]$. Then the following assertions are equivalent:

(a) $-A$ generates a sectorially contractive analytic C_0-semigroup

(of angle θ).
(b) $-\Sigma_\theta \subset \rho(A)$ and $\|R(A,\zeta)\| \le |\zeta|^{-1}$ for all $\zeta \in -\Sigma(\theta)$.

Recall from section 10 that linear operators associated with quasi-coercive forms are m-accretive, and in this case these operators are called regularly accretive. Rather similar, one of the main results in this section consists of the fact that quasi-coercive sesquilinear forms give rise to analytic semigroups and their (sectorial) infinitesimal generators are the operators associated with these forms:

THEOREM 11.4. The operator $A_\mathfrak{a}$ associated with a quasi-coercive form $\mathfrak{a}$ in a Hilbert space triple (V, H, V_a^*) is sectorial.

In fact, let $\mathfrak{a}$ be a coercive form with lower bound θ, that is, $\operatorname{Re}\mathfrak{a}(u,u) \ge \theta\,\|u\|_V^2$, and let $\lambda \in \mathbb{C}$ such that $\operatorname{Re}\lambda \ge 0$. Then the form $\mathfrak{a}(u,v) + \lambda\langle u,v\rangle$ is also coercive, and the associated operator $A+\lambda : \mathfrak{D}(A) \to H$ proves to be bijective. This implies that for $f \in H$ there is $u \in \mathfrak{D}(A)$ such that $Au + \lambda u = f$ and

$$\mathfrak{a}_\lambda(u,v) \;:=\; \mathfrak{a}(u,v) + \lambda\langle u,v\rangle_H \;=\; \langle f,v\rangle_H.$$

Letting $v = u$, we especially have

$$\mathfrak{a}_\lambda(u,u) \;=\; \mathfrak{a}(u,u) + \lambda\,\|u\|_H^2 \;=\; \langle f,u\rangle_H.$$

A first estimation gives

$$|\lambda|\;\|u\|_H^2 \le \Theta\,\|u\|_V^2 + \|f\|_H\,\|u\|_H\,,$$

while a second one, using the coercivity estimation, yields

$$\theta\,\|u\|_V^2 \le \mathfrak{a}(u,u) \;=\; \langle f,u\rangle_H - \lambda\,\|u\|_H^2 \le \|f\|_H\,\|u\|_H\,.$$

Combining both estimates, we obtain

$$|\lambda|\;\|u\|_H^2 \le (1 + \frac{\Theta}{\theta})\,\|f\|_H\,\|u\|_H\,,$$

hence

$$\|\lambda(A+\lambda)^{-1}f\| \le (1 + \frac{\Theta}{\theta})\,\|f\|$$

and $\|\lambda(A+\lambda)^{-1}\| \le M := 1+\Theta/\theta$ for all $\lambda \in \mathbb{C}$ with $\operatorname{Re}\lambda \ge 0$. This shows that the suffcient criterion for sectorialness of A as quoted in theorem 11.1 is fulfilled.

12. Applications to Linear Differential Operators

This final section deals with applications of abstract functional analysis to partial differential operators and the problems associated with them. We want to start with the concept of *linear partial differential operator* and continue with the consideration of specific problems associated with this kind of operators.

Remember first the notation of *multi-index*, which is nothing but a d-tuple $\alpha = (\alpha_1, \ldots, \alpha_d)$ with values $\alpha_i \in \mathbb{N}_0$ and with modulus $|\alpha| := \alpha_1 + \ldots + \alpha_d$. Multi-indices are used to shorten multiple partial derivatives $\partial^\alpha f := \partial_1^{\alpha_1} \partial_2^{\alpha_2} \ldots \partial_d^{\alpha_d} f$ of a sufficiently often differentiable and complex-valued function f, but also to denote the multiple powers $\xi^\alpha := \xi_1^{\alpha_1} \cdot \xi_2^{\alpha_2} \cdot \ldots \cdot \xi_d^{\alpha_d}$ of a vector $\xi = (\xi_1, \xi_2, \ldots, \xi_d) \in \mathbb{C}^d$.

By the usage of multi-indices, a *linear partial differential operator* $\mathcal{A}$ of even order $2m$ is defined by

$$\mathcal{A}(x, \cdot\,) := \sum_{|\alpha| \le 2m} c_\alpha(x)\, \partial^\alpha \tag{12.1}$$

for $x \in \overline{G}$, the closure of a *domain* G, that is, an open and connected subset of an Euclidean space $\mathbb{R}^d$. This domain may be bounded or unbounded and even the whole of $\mathbb{R}^d$.

We shall call $\mathcal{A}$ *essentially real*, if the coefficients c_α of highest order $|\alpha| = 2m$ are real-valued. Moreover, the expression $\mathcal{A}$ is said to be an *operator with constant coefficients*, if all functions c_α are constant in $\overline{G}$.

In case of $G = \mathbb{R}^d$ respectively $\partial G = \emptyset$, where $\partial G := \overline{G} \setminus \mathring{G}$ is the boundary of G, we denote the linear partial differential equation $\mathcal{A}u = f$ with $\mathcal{A}$ given by (12.1) a *free space problem.* On the other hand, partial differential equations considered in domains with non-empty boundary are usually furnished with *boundary conditions*, which fix possible solutions and their normal derivatives up to a certain order in the boundary ∂G. Thus, when having in mind linear equations of the form $\mathcal{A}u = f$ with imposed boundary conditions, we shall speak of *linear boundary value problems.*

The most general form of a boundary value problem posed in a domain $G \subsetneq \mathbb{R}^d$ is of the form

$$\left\{ \begin{array}{rcll} \mathcal{A}(x,u) &=& f(x) & (x \in G), \\ \mathcal{B}_k(x,u) &=& g_k(x) & (x \in \partial G \text{ and } k = 0, \ldots, m-1), \end{array} \right. \tag{12.2}$$

where $\mathcal{A}$ is a linear differential operator of order $2m$ and $\mathcal{B}_0, \ldots, \mathcal{B}_{m-1}$ are *boundary operators* of order m_k, each of them defined by

$$\mathcal{B}_k(x, u) \;=\; \sum_{|\alpha| \le m_k} b_{k,\alpha}(x)\, \partial^\alpha u(x).$$

To introduce the boundary operators $\mathcal{B}_k$, it is sufficent that the real or complex-valued functions $b_{k,\alpha}$ are defined in a neighbourhood of the boundary ∂G only. Then each operator $\mathcal{B}_k$ is a linear differential operator in its own right of order $m_k \le 2m-1$, which maps the function $u : \overline{G} \to \mathbb{C}$ to a function $\mathcal{B}_k u : \partial G \to \mathbb{C}$ belonging to a so-called *boundary space* $B_k(\partial G)$.

The collection $\{B_k\}_{k=0}^{m-1}$ of boundary operators in (12.2) is said to be *homogeneous* if $\mathcal{B}_k(x, \cdot) = 0$ for $x \in \partial G$ and $k = 0, \ldots, m-1$. Moreover, it is denoted *normal* if

(B-1) $m_i \neq m_j$ for $i \neq j$,

(B-2) for all $\xi \neq 0$ being normally positioned in $x \in \partial G$ we have

$$\sum_{|\alpha| = m_k} b_{k,\alpha}(x)\, \xi^\alpha \neq 0 \qquad (k = 0, \ldots, m-1).$$

Finally, our collection of boundary operators is called a *Dirichlet system*, if it is normal and the values m_k form a permutation of the set $\{0, \ldots, m-1\}$, so one may reorder the operators $\mathcal{B}_k$ to obtain $m_k = k$ $(0 \le k \le m-1)$. The most regarded set of boundary conditions in applications is the *homogeneous Dirichlet system* defined by

$$\mathcal{B}_k(x, u) \;:=\; (\partial_\nu^k u)(x) \;=\; 0 \tag{12.3}$$

for $x \in \partial G$ and $k = 0, \ldots, m-1$, where $(\partial_\nu^k u)(x)$ is the k-th derivative of a function u along the outer normal vector ν in $x \in \partial G$.

If the set of boundary conditions in (12.2) is a Dirichlet system, then we denote a solution $u : \overline{G} \to \mathbb{C}$ of this problem *classical*, if

(a) $u \in C^{2m}(G) \cap C^{m-1}(\overline{G})$,

(b) u fullfills $\mathcal{A}(x, u) = f(x)$ in each $x \in G$ and

(c) the $m-1$ equations $\mathcal{B}_k(x, u) = g_k(x)$ hold in every $x \in \partial G$.

Especially to each *second-order* boundary value problem there is assigned exactly one boundary operator $\mathcal{B}_0$, hence problem (12.2) becomes into

$$\left\{ \begin{array}{ll} \mathcal{A}(x, u) = f(x) & (x \in G), \\ \mathcal{B}_0(x, u) = g_0(x) & (x \in \partial G). \end{array} \right. \tag{12.4}$$

Boundary value problems of the form (12.4) and governed by a linear differential operator $\mathcal{A}$ of second order are obtained by various boundary operators $\mathcal{B}_0$. We denote

(BC-1) $\mathcal{B}_0(x,u) = u(x)$ for $x \in \partial G$ a *Dirichlet condition*,

(BC-2) $\mathcal{B}_0(x,u) = u_\nu(x)$ for $x \in \partial G$ a *Neumann condition*,

(BC-3) $\mathcal{B}_0(x,u) = \alpha(x)u(x) + \beta(x)u_\nu(x) = 1$ for $x \in \partial G$ a *Robin condition*, where the functions α, β must fulfill

$$\alpha(x)^2 + \beta(x)^2 \neq 0 \qquad (x \in \partial G).$$

Due to physical reasons, one often has to split the boundary set ∂G into disjoint subsets $\Gamma_1, \Gamma_2, \Gamma_3$, on which the conditions (BC-1), (BC-2) and (BC-3) are prescribed. We then shall speak of *mixed* boundary value problems for the second-order operator $\mathcal{A}$.

The concept of *ellipticity* has been shown to be helpful in order to classify linear partial differential expressions by a simple algebraic criterion. Let $\mathcal{A}$ be a linear differential operator of order $2m$ be given by (12.1) and let

$$p(x,\xi) := \sum_{|\alpha|=2m} c_\alpha(x)\, \xi^\alpha \quad \text{for } \xi = (\xi_1, \dots, \xi_d) \in \mathbb{R}^d$$

be the *characteristic function* or the *main symbol* of $\mathcal{A}$, which is built up by the summands of highest order in (12.1).

(E-1) The expression $\mathcal{A}$ is called *elliptic* in $x \in \overline{G}$, if

$$|p(x,\xi)| \neq 0 \quad \text{for all } \xi \neq 0,$$

and $\mathcal{A}$ is said to be *uniformly elliptic* in G (in $\overline{G}$), if it is elliptic in *each* point $x \in G$ (in $x \in \overline{G}$). We then call a boundary value problem $\mathcal{A}u = f$ *elliptic*, if $\mathcal{A}$ is so.

(E-2) $\mathcal{A}$ is called *properly elliptic* in $\overline{G}$, if it is elliptic and if for each fixed $x \in \partial G$ and all linear independent vectors $\xi, \eta \in \mathbb{R}^d$ the main symbol $P_{\mathcal{A}}(x, \xi + \tau\eta)$ has, with respect to the variable τ, exactly m roots with positive imaginary part and exactly m roots with negative imaginary part. It can be shown that if $\mathcal{A}$ is *essentially real* or $G \subset \mathbb{R}^d$ with $d \geq 3$, then any elliptic differential operator is also properly elliptic.

(E-3) The concept of ellipticity is often not good enough to obtain satisfying results. Hence the expression $\mathcal{A}$ of order $2m$ is called *strongly elliptic* in $x \in \overline{G}$, if there is $\theta_x > 0$ such that

$$\operatorname{Re}(-1)^m\, p(x,\xi) \;\geq\; \theta_x \cdot |\xi|^{2m} \qquad \text{for all } \xi \neq 0.$$

(E-4) $\mathcal{A}$ is called *uniformly strongly elliptic* in $\overline{G}$, if there is $\theta > 0$ such that

$$\operatorname{Re}(-1)^m\, p(x,\xi) \;\geq\; \theta \cdot |\xi|^{2m} \qquad \text{for all } \xi \neq 0 \text{ and } x \in \overline{G}.$$

The most well-known linear partial differential operator of second order is the *Laplacian*

$$\Delta = \partial^2/\partial {x_1}^2 + \ldots + \partial^2/\partial {x_d}^2,$$

whose negative version $-\Delta$ constitutes the prototype of what is meant by an uniformly strongly elliptic operator. One easily computes $\operatorname{Re} -p_{-\Delta}(x,\xi) = |\xi|^2$ for $\xi \in \mathbb{R}^d$, hence the expression $-\Delta$ often serves as a first example when studying linear-elliptic operators.

Besides the class of linear-elliptic operators there are two further important classes of linear partial differential operators. They are called to be of *parabolic* and *hyperbolic* type. In order to introduce these concepts, we consider differential expressions on cylindrical subsets $I \times G$, where $I = (0,\tau) \subset \mathbb{R}$ and $G \subset \mathbb{R}^d$ is a domain. Then a *parabolic* differential operator $\mathcal{A}$ is one of the form

$$\mathcal{A}(t,x,u) \;=\; \frac{\partial u}{\partial t}(x) \,+\, \mathcal{A}_1(x,u),$$

where $\mathcal{A}_1$ is *uniformly elliptic* in G, and $\mathcal{A}$ is of *hyperbolic* type, if it is of the form

$$\mathcal{A}(t,x,u) \;=\; \frac{\partial^2 u}{\partial t^2}(x) \,+\, \mathcal{A}_1(x,u),$$

where again $\mathcal{A}_1$ is uniformly elliptic in G. Simple examples of parabolic and hyperbolic operators are the *heat operator* $\partial_t - \Delta$ and the *wave operator* $\partial_{tt} - \Delta$, respectively. With these operators we associate the linear evolution equations

$$\frac{\partial u}{\partial t} + \mathcal{A}(\cdot,u) = f \qquad \text{and} \qquad \frac{\partial^2 u}{\partial t^2} + \mathcal{A}(\cdot,u) = f,$$

both of them called *initial value problems*, when considered with the prescribed start value $u(0) = u_0$ (and in addition $\dot{u}(0) = u_1$ in the hyperbolic case).

When considering parabolic or hyperbolic equations, we speak of so-called *initial-boundary value problems*, if for every time $t \in (0,T]$ the solution fulfills a certain set of boundary conditions in every

$x \in \partial G$, where we especially have in mind the homogeneous Dirichlet boundary conditions (12.3). However, in case of $G = \mathbb{R}^d$, we obtain free space problems or pure initial value problems for the differential operators of parabolic or hyperbolic type.

It is the aim of this section to apply all the presented functional analytic tools to the just enumerated problems. Let $\mathcal{A}$ be a linear partial differential operator of order $k \in \mathbb{N}$ (not necessarily even) and given by

$$\mathcal{A}(x,u) := \sum_{|\alpha| \le k} c_\alpha(x)\, \partial^\alpha u. \tag{12.5}$$

Then by a *realization* of $\mathcal{A}$ in a Banach space $\mathbb{E}(G)$ of functions $u : G \to \mathbb{C}$ is meant any linear operator

$$A : \mathfrak{D}(A) \subset \mathbb{E}(G) \to \mathbb{E}(G)$$

with domain of definition $\mathfrak{D}(A)$. The Banach space $\mathbb{E}(G)$ should contain the space $C_c^\infty(G)$ of test functions[2] as a dense linear subspace. We also require for each $u \in \mathfrak{D}(A)$ that first $\mathcal{A}u$ is well-defined by $Au = \mathcal{A}u$ and second $\mathcal{A}u \in \mathbb{E}(G)$ holds. Clearly, our first condition means that the set $\mathfrak{D}(A)$ may contain only functions $u : G \to \mathbb{C}$ being sufficiently often differentiable in some sense. But $\mathfrak{D}(A)$ should also include the space $C_c^\infty(G)$ of test functions, so $\mathfrak{D}(A)$ lies dense in $\mathbb{E}(G)$ as well and A appears as a densely defined operator in $\mathbb{E}$.

In the second part of this section we shall point out that

- for a linear partial differential expression $\mathcal{A}$ there can be defined a couple of densely defined and closable linear operators A acting either in the Hölder spaces $C^{k,\gamma}(G)$ ($k \in \mathbb{N}_0$, $0 < \gamma \le 1$) or in the Lebesgue spaces $L^p(G)$ ($1 \le p < \infty$);
- using elliptic estimates, the realization of a strongly elliptic differentiable operator acting in Sobolev spaces under homogeneous Dirichlet boundary conditions is closed;
- the study on realizations of differential operators can be supported by that on their adjoints, where those are determined by so-called

[2] The space $C_c^\infty(G)$ consists of indefinitely differentiable functions having *compact* support

$$\operatorname{supp}(\phi) := \overline{\{x \in G : \phi(x) \neq 0\}},$$

usually denoted the *space of test functions.* This space plays an outstanding role in analysis, since it lies dense in a multiplicity of other functional spaces.

adjointness theorems;

- by starting with quasi-coercive sesquilinear forms, created by strongly elliptic differential expressions, we can obtain realizations in $L^2(G)$, which prove to be densely defined and closed;
- weak solutions of initial-boundary value problems exist and are unique, supposed the governing sesquilinear form is the Dirichlet form of a strongly elliptic differential operator;
- the realizations of a regularly elliptic expression $\mathcal{A}$ in $L^2(G)$ are Fredholm, so the alternative theorem (5.6) may be applied to the equation $\mathcal{A}\,u = f$;
- strongly elliptic expressions give rise to m-accretive and sectorial realizations in $L^2(G)$, so the related parabolic and hyperbolic initial value problem prove to be well-posed;
- formally self-adjoint and half-bounded differential expressions lead to self-adjoint realizations in $L^2(G)$;

This enumeration is of course only a small selection within the totality of applications of functional analytic methods to linear partial differential operators and related equations, but may give a good impression of their capabilities.

I. Realizations in the spaces $C^\gamma(G)$ and $L^p(G)$. We basically want to distinguish between two classes of functional spaces, in which realizations of a linear differential operator $\mathcal{A}$ are studied:

1. The notation of classical derivative $\partial^\alpha u$, applied to a function $u: G \to \mathbb{C}$, leads us to the usage of the spaces $C^\gamma(G)$ of *Hölder-constinuous* functions, which constitute Banach spaces when endowed with a suitable norm.

   Preliminary, a function $u: G \to \mathbb{K}$ is called *Hölder-continuous* with respect to the parameter $\gamma \in (0,1]$, if its *Hölder-norm* is finite, that is,

$$|u|_\gamma := \sup_{x,y\,\in\, G,\, x\neq y} \frac{|u(x) - u(y)|}{|x-y|^\gamma} < \infty.$$

   Let $C^{k,\gamma}(\overline{G})$ be the linear space of functions $u \in C^k(\overline{G})$, whose partial derivatives $\partial^\alpha u$ are Hölder-continuous for all $|\alpha| = k$. Then the *Hölder norm* $|u|_{k,\gamma}$ of a function $u \in C^{k,\gamma}(\overline{G})$ is defined by the sum of the norm $\|u\|_k := \sum_{|\alpha|\leq k} \sup_{x\,\in\,\overline{G}} |\partial^\alpha u(x)|$ and the Hölder norms of the partial derivatives $\partial^\alpha u$ for $|\alpha| = k$:

$$|u|_{k,\gamma} := \|u\|_k + \sum_{|\alpha|=k} |\,\partial^\alpha u|_\gamma.$$

The spaces $C^{k,\gamma}(\overline{G})$ prove to be complete with respect to this norm, but they are not reflexive.

A realization of a differential operator $\mathcal{A}$ of order $2m$ acting in Hölder space $C^\gamma(G)$ is nothing but a mapping

$$A : \mathfrak{D}(A) \subset C^\gamma(G) \to C^\gamma(G),$$

where $\mathfrak{D}(A)$ forms a linear subspace of $C^{2m,\gamma}(G)$ restricted by boundary conditions, for example by a Dirichlet system.

There is an extensive theory on the realizations of linear partial differential operators acting in Hölder spaces, which provide nice results especially for operators of second order. We should mention J. SCHAUDER as the outstanding pioneer in the investigation of this subject. But SCHAUDER uses classical methods for the setup of his theory, coming from potential theory and complex analysis, so we do not want to go into the details (a good reference for this subject is [6]).

2. Instead of classical derivatives, the modern theory of partial differential equations is based on the concept of *weak derivative*, since this approach leads to larger spaces of differentiable functions than the already mentioned Hölder spaces, usually denoted as *Lebesgue-Sobolev spaces.*

   First recall that for any domain $G \subset \mathbb{R}^d$ the Hilbert space $L^2(G)$ is a complete normed space consisting of measurable functions $f : G \to \mathbb{C}$ such that

$$\int_G |f|^2 \, dx < \infty,$$

   where the inner product in $L^2(G)$ is defined by

$$\langle f, g \rangle := \int_G f \cdot \overline{g} \, dx \quad \text{for } f, g \in L^2(G).$$

   This space is a specific member in the scale of the so-called *Lebesgue spaces* $L^p(G)$ $(1 \le p < \infty)$, whose norms are defined by

$$|f|_p := \Big(\int_G |f|^p \, dx\Big)^{\frac{1}{p}} < \infty$$

   for equivalence classes of measurable functions $f \in L^p(G)$.

The space $L^\infty(G)$, however, consists of those equivalence classes of measurable functions $f:G\to\mathbb{C}$, for which

$$\|f\|_\infty := \inf\{\, a\geq 0 : |f|\leq a \ \mu\text{-almost everywhere}\,\} < \infty.$$

We next turn to the notation of *weak derivative.* Preliminary, any *locally integrable function*[3] $u : G\to\mathbb{C}$ is said to be *weakly differentiable* with respect to a given multi-index α, if there is a further locally integrable function v such that

$$\int_G v(x)\,\phi(x)\,dx = (-1)^{|\alpha|}\int_G u(x)\,\partial^\alpha\phi(x)\,dx \text{ for all } \phi\in C_c^\infty(G).$$

We denote $\partial^\alpha u := v$ as the *α-weak derivative* of u in this case. The function u is said to be *k-times weakly differentiable*, if u has α-weak derivatives for every index α with $|\alpha|\leq k$. Taking weak derivatives of locally integrable functions with respect to the multi-index α arises as a linear mapping

$$\partial^\alpha : D_\alpha\subset L^1_{loc}(G)\to L^1_{loc}(G),$$

where D_α forms a linear subspace of $L^1_{loc}(G)$. Let us also notice that the concept of weak derivative is compliant with that of the classical derivative: if a function u is classical differentiable in G with classical derivative $\partial^\alpha u$, then u is also weakly differentiable, and its weak derivative is equal to $\partial^\alpha u$.

The space $W^{k,p}(G)$ $(1\leq p<\infty)$ is the *Lebesgue-Sobolev space*

$$W^{k,p}(G) := \{\, f\in L^p(G) : \partial^\alpha f\in L^p(G),\ |\alpha|\leq k\,\}.$$

If $p=2$, the space $H^k(G) := W^{k,2}(G)$ can be furnished with inner product

$$\langle f,g\rangle_k := \sum_{|\alpha|\leq k}\int_G \partial^\alpha f\cdot\partial^\alpha\overline{g}\,dx\,, \qquad f,g\in H^k(G)$$

and norm $|f|_k = \langle f,f\rangle^{1/2}$, so $H^k(G)$ forms a Hilbert space in its own right. In other words, the space $H^k(G)$ consists of square-integrable functions, which are at least k-times weakly differentiable and all these weak derivatives belong to $L^2(G)$ again.

We also want to consider the linear subspace $W_0^{k,p}(G)$ of $W^{k,p}(G)$ fulfilling homogeneous Dirichlet boundary conditions. This space consists of those functions $u\in W^{k,p}(G)$, which are ap-

[3] A function $u:G\to\mathbb{C}$ is said to be *locally integrable* and to belong to the space $L^1_{loc}(G)$, if u is Lebesgue-integrable on each compact subset of G.

proximated by a sequence $\{\phi_n\}_{n\in\mathbb{N}}$ of *test functions* with respect to the norm $|\cdot|_p$, that is, each ϕ_n belongs to the space $C_c^\infty(G)$ and $|u-\phi_n|_p\to 0$. Clearly, since $W_0^{k,p}(G)$ is a closed subspace of $W^{k,p}(G)$, it forms a Banach space again. Furthermore, functions belonging to $W_0^{k,p}(G)$ fulfill the k boundary conditions of a homogeneous Dirichlet system. For $p=2$, however, the space $H_0^k(G)$ owns the inner product and norm as defined in the larger space $H^k(G)$, so it is a Hilbert space as well.

Finally, a realization of a linear differential operator $\mathcal{A}$ of order k in Lebesgue space $L^p(G)$ is any linear mapping

$$A:\mathfrak{D}(A)\subset L^p(G)\to L^p(G),$$

for which $\mathfrak{D}(A)$ forms a linear subspace of $W^{k,p}(G)$, for example restricted by homogeneous Dirichlet boundary conditions.

II. The closeness of strongly elliptic Dirichlet operators. One often has to deal with the realization A_D of a strongly elliptic differential operator $\mathcal{A}$ in $L^p(G)$ having order $2m$ and domain of definition

$$\mathfrak{D}(A_D) := \{u\in W_0^{m,p}(G):\mathcal{A}u\in L^p(G)\},$$

so the functions in $\mathfrak{D}(A_D)$ satisfy all the m boundary conditions of the homogeneous Dirichlet system (12.3). By this reason it is reasonable to denote the realization A_D a *Dirichlet operator.*

We turn to the question of closeness of the realization A_D. Let the domain G and the expression $\mathcal{A}$ fulfill the following conditions:

(I) G is bounded in $\mathbb{R}^d$,

(I') G satisfies the m-th *extension property*, that is, the linear space $R_0(G)$ consisting of those functions $\phi\in W^{m,p}(G)$, which are restrictions $\phi|_G$ of functions $\phi\in C_c^\infty(\mathbb{R}^d)$, is dense in $W^{m,p}(G)$;

(II) $\mathcal{A}$ is uniformly strongly elliptic in G;

(III) the coefficient functions $a_{\alpha\beta}$ are essentially bounded in G, that is, $a_{\alpha\beta}\in L^\infty(G)$;

(III') the coefficient functions $a_{\alpha\beta}$ of highest order $|\alpha|+|\beta|=2m$ are uniformly continuous in the closure $\overline{G}$ of G.

Then, by showing the validness of the *elliptic estimate*

$$|u|_{2m}\ \le\ c\cdot\{\,|u|_p+|A_Du|_p\,\}$$

for $u\in\mathfrak{D}(A_D)$, where $|u|_p=\|u\|$ is the norm of u in $L^p(G)$, theorem 2.2 implies that the operator A_D is closed. Moreover, A_D is injective and the range of A_D is a closed subspace in $L^p(G)$.

III. Formally adjoint expressions and adjointness theorems. As we have already mentioned, it is often helpful to combine the study of a linear operator with that of its formally adjoint. Supposed the linear partial differential operator $\mathcal{A}$ of order k has coefficients $c_\alpha \in C^{|\alpha|}(G)$. We then find for $u \in C_c^\infty(G)$ and $\psi \in C_c^\infty(G)$ by repeated partial integration

$$\int_G \sum_{|\alpha|\le k} (c_\alpha\, \partial^\alpha u)\cdot \overline{\psi}\, dx \;=\; \int_G u \cdot \sum_{|\alpha|\le k} (-1)^{|\alpha|}\, \overline{\partial^\alpha(\overline{c_\alpha}\,\psi)}\, dx.$$

Hence there is a *formally adjoint* differential expression

$$\mathcal{A}^\dagger(x,u) \;:=\; \sum_{|\alpha|\le k} (-1)^{|\alpha|} \partial^\alpha(\overline{c_\alpha(x)}\, u(x)) \tag{12.6}$$

such that the identity

$$\int_G \mathcal{A}\phi\,\overline{\psi}\, dx \;=\; \int_G \phi\, \overline{\mathcal{A}^\dagger\,\psi}\, dx$$

holds for $\phi, \psi \in C_c^\infty(G)$, briefly $\langle \mathcal{A}\phi, \psi\rangle = \langle \phi, \mathcal{A}^\dagger\,\psi\rangle$ by means of the inner product in $C_c^\infty(G)$. Using LEIBNIZ's rule to compute the terms $\partial^\alpha(\overline{c_\alpha}\,u)$ in (12.6), there are functions $c'_\alpha : G \to \mathbb{C}$ such that

$$\mathcal{A}^\dagger(x,u) \;=\; \sum_{|\alpha|\le k} c'_\alpha(x)\; \partial^\alpha u(x).$$

Thus $\mathcal{A}^\dagger$ is uniquely determined by $\mathcal{A}$ and constitutes a differential expression of type (12.5), too. It may happen that $\mathcal{A}^\dagger = \mathcal{A}$ holds, and in this case we shall call the expression $\mathcal{A}$ *formally self-adjoint.*

A useful criterion to ensure closability of a linear operator A acting in Hilbert space $L^2(G)$ and induced by a differential expression $\mathcal{A}$ of order k is given by the following theorem (see also [17], p. 78), whose proof uses the presence of the formally adjoint expression $\mathcal{A}^\dagger$:

THEOREM 12.1. Let the expression $\mathcal{A}$ of order n be given by (12.5) with coefficients $c_\alpha \in C^k(G)$ for $|\alpha| \le k$ and the linear operator

$$A : \begin{cases} \mathfrak{D}(A) \;:=\; \{u \in C^k(G) \cap L^2(G) : \mathcal{A}\,u \in L^2(G)\}, \\ Au \;:=\; \mathcal{A}\,u \;\text{ for } u \in \mathfrak{D}(A) \end{cases}$$

be a realization of $\mathcal{A}$ in $L^2(G)$. Then A is closable.

Note that the domain of definition $\mathfrak{D}(A)$ of the operator A considered in theorem 12.1 is not restricted to any boundary conditions. It should also be mentioned that rather the closability of the differential operator A with domain of definition $\mathfrak{D}(A)$ like in theorem 12.1 is ensured, it is not an easy task to give an exact description for the domain of definition of its closure. But if the coefficients of $\mathcal{A}$ are constant, then the set $\mathfrak{D}(\overline{A})$ can be determined and coincides with the Hilbert-Sobolev space $H^k(G)$.

The closure $A_{min} := \overline{A_0}$ of a closable linear differential operator A_0 with domain of definition $\mathfrak{D}(A_0) = C_c^\infty(G)$ is denoted the *minimal realization* or the *minimal operator* of $\mathcal{A}$. The *maximal operator* A_{max} of $\mathcal{A}$, however, is by definition the adjoint of the minimal operator of $\mathcal{A}^\dagger$ with domain of definition $\mathfrak{D}({A^\dagger}_0) = C_c^\infty(G)$, that is, $A_{max} := ({A^\dagger}_{min})^*$. One always has $A_{min} \prec A_{max}$, and often any operator A fulfilling

$$A_{min} \prec A \prec A_{max}$$

is called a *realization* of $\mathcal{A}$, in opposite to our selected definition of this concept. Using this definition, we also want to consider the linear operators A_{min} and A_{max} as realizations of $\mathcal{A}$, too.

To the end, the question arises whether any realization of the formally adjoint expression $\mathcal{A}^\dagger$ is in relationship with the adjoint of some realization of $\mathcal{A}$. This problem is examined in [3] (section 3: Adjoints) even in the context of L^p-spaces. BROWDER's result for Hilbert spaces $L^2(G)$ and the Dirichlet realization A_D of a regularly elliptic differential expression $\mathcal{A}$ sounds as follows (corresponding to theorem 5, see p. 57):

THEOREM 12.2. Let the $2m$-order expression $\mathcal{A}$ be strongly elliptic with sufficiently smooth coefficients and defined on a smooth domain $G \subset \mathbb{R}^d$ such that the formal adjoint $\mathcal{A}^\dagger$ exists and its coefficients are uniformly bounded in G.

If A and $A^\dagger$ are the realizations of $\mathcal{A}$ and $\mathcal{A}^\dagger$ in $L^2(G)$ with homogeneous Dirichlet boundary conditions and domains of definition

$$\mathfrak{D}(A) = \mathfrak{D}(A^\dagger) := H^{2m}(G) \cap H_0^m(G),$$

then $A^\dagger$ is the Hilbert adjoint of A, that is, $A^* = A^\dagger$.

A similar result holds in the setting of the spaces $L^p(G)$ for $1 < p < \infty$. As it is quoted in [12], lemma 7.3.4, we have ${A_p}^* = {A^\dagger}_q$, where $q = p/(p-1)$ respectively $\frac{1}{p}+\frac{1}{q}=1$ and both operators

$$\begin{cases} \mathfrak{D}(A_p) := W^{2m,p}(G) \cap W_0^{m,p}(G), \\ A_p u := \mathcal{A} u \text{ for } u \in \mathfrak{D}(A_p), \end{cases}$$

$$\begin{cases} \mathfrak{D}(A^\dagger{}_q) := W^{2m,q}(G) \cap W_0^{m,q}(G), \\ A^\dagger{}_q u := \mathcal{A}^\dagger u \text{ for } u \in \mathfrak{D}(A^\dagger{}_q). \end{cases}$$

are densely defined and closed.

IV. Gårding's inequality and quasi-coercive Dirichlet forms. We next want to illustrate the usage of abstract sesquilinear forms for the analysis of linear equations of the form $\mathcal{A} u = f$, where $\mathcal{A}$ is a linear partial differential operator of order $2m$ and u, f are real or complex-valued mappings defined on a domain $G \subset \mathbb{R}^d$.

Preliminary, a linear differential expression $\mathcal{A}$ of order $2m$ is said to be in *divergence form*, if there are functions $a_{\alpha\beta} \in C^{|\alpha|}(G)$ for $|\alpha|, |\beta| \leq m$ such that $\mathcal{A}$ can be written as

$$\mathcal{A} := \sum_{|\alpha|\leq m, |\beta| \leq m} (-1)^{|\alpha|}\, \partial^\alpha (\, a_{\alpha\beta}(x)\, D^\beta). \tag{12.7}$$

In opposite to this definition let us call an arbitary differential expression given by (12.5) to be *in non-divergence form.*

It is a matter of fact that every expression $\mathcal{A}$ in divergence form can be transformed into an expression $\mathcal{A}'$ in *non-divergence form.* Reversely, a non-divergence form expression $\mathcal{A}'$ given by (12.5) can be transformed into an expression $\mathcal{A}$ in divergence form (12.7), if $c_\alpha \in C^{2m-|\alpha|}(G)$ for $|\alpha| \leq 2m$, that is, if the coefficients are sufficiently many differentiable in G. Comparing the expressions (12.5) and (12.7), it turns out that $\mathcal{A}'$ is strongly elliptic in G if and only if $\mathcal{A}$ is so.

To show the purpose of a linear differential operator being in divergence form, let the space $C_c^\infty(G)$ of test functions be furnished with the inner product

$$\langle \phi, \psi \rangle := \int_G \phi\, \overline{\psi}\, dx \quad \text{for all } \phi, \psi \in C_c^\infty(G).$$

Now, applying $\mathcal{A}$ given in divergence form to $\phi \in C_c^\infty(G)$, then multiplying $\mathcal{A}\phi$ with $\overline{\psi} \in C_c^\infty(G)$ and finally performing m-times partial integration leads to

$$B_{\mathcal{A}}(\phi,\psi) := \langle \mathcal{A}\,\phi,\psi\rangle = \sum_{|\alpha|\le m,|\beta|\le m} \int_G a_{\alpha\beta}\,\partial^\alpha\phi\, D^\beta\overline{\psi}\,dx,$$

since all the boundary terms vanish. The mapping

$$B_{\mathcal{A}} := C_c^\infty(G)\times C_c^\infty(G)\to\mathbb{C}$$

is a sesquilinear form, called the *Dirichlet form* of $\mathcal{A}$ in G. If the coefficent functions $a_{\alpha\beta}$ of $\mathcal{A}$ are measurable and essentially bounded, then the mapping $B_{\mathcal{A}}$ is continuous and thus can continuously be extended to the product space $H_0^m(G)\times H_0^m(G)$.

We are mainly interested in a $2m$-order linear differential operator $\mathcal{A}$ given in divergence form, which together with the underlying domain G fulfills the already mentioned conditions (I), (I'), (II), (III) and (III'). GÅRDING*'s inequality* can be seen as a bridge from the theory of linear-elliptic differential operators to abstract functional analysis:

THEOREM 12.3. Let the domain G fulfill (I), (I') and let the linear differential expression $\mathcal{A}$ of order $2m$ fulfill (II), (III), (III').
Then there is $\lambda_0\in\mathbb{R}$ and $a_0>0$ such that for all $u\in H_0^m(G)$

$$B_{\mathcal{A}}(u,u)+\lambda_0\,\|u\|^2 = \langle\,(\mathcal{A}+\lambda_0)\,u,u\rangle \ge a_0\,|u|_m{}^2. \tag{12.8}$$

In other words, theorem 12.3 quotes that the Dirichlet form associated with the operator $\mathcal{A}+\lambda_0$ is coercive, hence the Lax-Milgram theorem may be applied. It should be clear that this assertion also holds for all $\lambda>\lambda_0$.

The most important class of linear partial differential operators in divergence form with respect to applications is that of *second order*, that is, our operator $\mathcal{A}$ is of the form

$$\mathcal{A}(x,\cdot) = -\sum_{i=1}^d \partial_i\,(\sum_{j=1}^d a_{ij}(x)) + \sum_{j=1}^d (b_j(x)\,\partial_j) + c(x), \tag{12.9}$$

which is nothing but (12.7) for $m=1$. The Dirichlet form $B_{\mathcal{A}}$ belonging to $\mathcal{A}$ is the sesquilinear form

$$B_{\mathcal{A}}(u,v) = \int_G (\sum_{i,j=1}^d a_{ij}\,\partial_i u\,\partial_j\overline{v} + \sum_{j=1}^d b_j\,\partial_j u\,\overline{v} + c\,u\,\overline{v})\,dx.$$

A short computation shows that if the function c is large enough in comparison with the sum of partial derivatives of the functions b_j, then $B_{\mathcal{A}}$ is coercive. More precisely, $B_{\mathcal{A}}$ proves to be coercive if

$$c(x) \geq \frac{1}{2} \sum_{k=1}^{d} \frac{\partial b_k}{\partial x_k}(x) \tag{12.10}$$

holds in all points $x \in G$.

On the other hand, if the matrix $A(x) = (a_{ij}(x))_{i,j=1}^{d}$ of second-order coefficients in (12.9) fails to be positive definite for all $x \in G'$, where $G' \subset G$ has positive measure, then $B_{\mathcal{A}}$ cannot be coercive on $H_0^1(G)$. Also, if $A(x)$ is positive definite in all $x \in G$ and condition (12.10) does not hold. then the functions b_j disturb the coercivity of $B_{\mathcal{A}}$. But after all, the form $B_{\mathcal{A}}$ proves at least to be quasi-coercive, which is nothing but the assertion of GÅRDING's inequality.

GÅRDING's inequality remains even true for the Dirichlet forms of second-order differential operators, if the basic conditions (I)' and (III') are dropped:

THEOREM 12.4. Let $G \subset \mathbb{R}^d$ fulfill (I) and the differential operator $\mathcal{A}$ of second order and given in divergence form fulfill (II) and (III). Then there is $\lambda_0 \in \mathbb{R}$ and $a_0 > 0$ such that

$$B_{\mathcal{A}}(u,u) + \lambda_0 \|u\|^2 = \langle (\mathcal{A} + \lambda_0)\, u, u \rangle \geq a_0 \, |u|_1^{\,2}$$

for all $u \in H_0^1(G)$, where a_0 is the uniform lower bound of the eigenvalues of the matrices $A(x)$ with $x \in G$.

V. Weak solutions of boundary and initial-boundary value problems. The variational approach to stationary and evolutional boundary value problems provides the concept of weak solution, and in this part of the section we want to apply the abstract theory boundary value problems governed by strictly elliptic differential operators by quoting the related existence and uniqueness assertions.

We basically assume that

(DP-1) $\mathcal{A}$ is a the differential operator of order $2m$, strongly elliptic in a *bounded* domain $G \subset \mathbb{R}^d$ and given in divergence form;

(DP-2) the coefficients $a_{\alpha\beta}$ belong to $L^\infty(G)$ and the top-order coefficients are uniformly continuous in $\overline{G}$.

It is our first aim to solve the *homogeneous Dirichlet problem*

$$\begin{cases} (\mathcal{A}u)(x) = f(x) & (x \in G), \\ u(x) = \partial u/\partial\nu(x) = \ldots = \partial u^{m-1}/\partial\nu^{m-1}(x) = 0 & (x \in \partial G). \end{cases}$$

Let $V = H_0^m(G)$ and $H = L^2(G)$, hence the pair (V, H) is a Hilbert triple. By GÅRDING's inequality, the Dirichlet form

$$B_{\mathcal{A}}(u,v) := \langle \mathcal{A}u, v\rangle \qquad (u, v \in V)$$

is quasi-coercive with some $\lambda_0 \in \mathbb{R}$. Then, by means of the Lax-Milgram theorem, we obtain the following *well-posedness theorem*:

THEOREM 12.5. For $\lambda \geq \lambda_0$, the variational equation

$$B_{\mathcal{A}}(u,v) + \lambda\langle u, v\rangle = f(v) \quad (v \in H_0^m(G))$$

owns for $f \in H^{-m}(G) := H_0^m(G)^*$ a unique solution $u \in H_0^m(G)$ depending continuously on the right-hand side f and called the *weak solution* of the homogeneous Dirichlet problem.

VI. The Fredholm property of elliptic operators. This statement is properly not correct, since it only proves to be true for properly elliptic differential operators defined in a *bounded* domain $G \subset \mathbb{R}^d$ and endowed with certain boundary conditions, for example those given by a homogeneous Dirichlet system.

We first want to present RELLICH*'s embedding theorem*, which holds for Sobolev spaces whose underlying domain is *bounded*. It can be seen as a key theorem in the analysis of linear-elliptic boundary problems on such domains:

THEOREM 12.6. If $G \subset \mathbb{R}^d$ is bounded and owns smooth boundary ∂G, then the continuous embeddings $\iota_k : H_0^k(G) \hookrightarrow L^2(G)$ are compact for all $k \in \mathbb{N}$.

The basic approach to investigate the Fredholm property of a properly elliptic operator $\mathcal{A}$ in divergence form and of order $2m$ consists of two small changes for the Dirichlet realization A_D having domain of definition $\mathfrak{D}(A) = H^{2m}(G) \cap H_0^m(G)$: let us consider the linear operator A_D as

(F-1) a mapping $A_D : H_0^m(G) \to L^2(G)$, which is continuous due to the essential boundedness of the coefficients $a_{\alpha,\beta}$ in (12.7);

(F-2) the sum $A_1 + A_2$ of two further linear operators, where we shall assume that $A_1 : H_0^m(G) \to L^2(G)$ is the main part having weak derivatives of order $2m$ only, whereas $A_2 : H_0^m(G) \to L^2(G)$ is the part of A_D consisting of weak derivatives with order less than $2m$.

Let λ_0 be the smallest parameter for the Dirichlet form $B_{A+\lambda_0}$ to be coercive. Since

$$A_D = (A_1 + \lambda_0) + (A_2 - \lambda_0)$$

and $A_2 - \lambda_0$ may be seen as a compact perturbation of $A_1 + \lambda_0$, which maps $H^m(G)$ one-to-one onto $L^2(G)$ and consequently is Fredholm with vanishing index, the operator A_D is Fredholm again with index $\chi(A_D) = 0$.

VII. Spectral theory and sectorialness of strongly elliptic operators. Let us consider again a strongly elliptic differential operator $\mathcal{A}$ given in divergence form and its Dirichlet realization A_D in a bounded domain $G \subset \mathbb{R}^d$. By Gårding's inequality, the Dirichlet form $B_{\mathcal{A}}$ is quasi-coercive with respect to the parameter λ_0, hence $A_D + \lambda$ proves to be bijective for all $\operatorname{Re}\lambda > \lambda_0$. Furthermore, the homogeneous Dirichlet problem in $L^2(G)$

$$A_D u + \lambda u = f \qquad (f \in L^2(G)) \tag{12.11}$$

is well-posed, which is a direct consequence of theorem 12.5.

In the special case, where $f \in L^2(G)$, the solutions u of (12.11) belong to the Hilbert-Sobolev space $H^{2m}(G)$ and are called *strong*. It should be clear that each strong solution of (12.11) is also a weak solution. But in order to show that these solutions u are even classical, that is, $u \in C^{2m}(G) \cap C^m(\overline{G})$, methods of *regularity theory* for elliptic problems have to be applied.

Let us turn to the spectrum of A_D. Since $-\Delta_D + \lambda$ forms a bijective and closed operator between the spaces $H^{2m}(G) \cap H_0^m(G)$ and $L^2(G)$ for each $\lambda \in C$ with $\operatorname{Re}\lambda \geq \lambda_0$, we surely have

$$\{\zeta \in \mathbb{C} : \operatorname{Re}\zeta \geq \lambda_0\} \subset \rho(A_D) \text{ and } \sigma(A_D) \subset \{\zeta \in \mathbb{C} : \operatorname{Re}\zeta < \lambda_0\}.$$

Furthermore, if G is bounded, then A_D is an operator with pure point spectrum.

We can also make assertions on the accretivity and sectorialness of the operator A_D. As theorem (10.5) (d), (e) shows, $A_D + \lambda_0$ is the generator of a semigroup of bounded linear operators that is strongly continuous and analytic at the same time. Hence the initial value problems for the heat and wave equation governed by the operator

$A_D + \lambda_0$ is well-posed. By perturbation theory of semigroups, this is also true for both problems governed by the operator A_D itself.

Let us finally consider the heat equation for the negative Laplacian $-\Delta_D$ in the entire space $\mathbb{R}^d$, which of course is a free space problem. Theorem 11.2 ensures that the realization $-\Delta_D$ in Hilbert space $L^2(\mathbb{R}^d)$ with domain of definition

$$\mathfrak{D}(-\Delta_D) \;=\; H^2(\mathbb{R}^d)$$

is the infinitesimal generator of a sectorially contractive analytic semigroup $\{W(t)\}_{t\geq 0}$, denoted the *Gauss-Weierstrass semigroup*. Hence there is a uniquely defined solution $u \in C^1(I \times \mathbb{R}^d, \mathbb{R})$ for the homogeneous Cauchy problem

$$\begin{cases} u_t - \Delta u \;=\; 0, \\ u(x,0) \;=\; u_0(x) \text{ for } x \in \mathbb{R}^d \end{cases}$$

with initial mapping $u_0 \in H^2_0(\mathbb{R}^d)$. Then one may represent the semigroup $\{W(t)\}_{t\geq 0}$ explicitly by means of the *heat kernel* function Φ_d, that is, for all $t > 0$ and $x \in \mathbb{R}^d$

$$(W(t)\, u_0)\,(x) = \frac{1}{(4\pi\, t)^{d/2}} \int_{\mathbb{R}^d} e^{-\frac{|x-y|^2}{4t}}\, u_0(y)\, dy = (\Phi_d(t) * u_0)\,(x).$$

VIII. Self-adjoint extensions of half-bounded operators. Remember the definition of the differential expression $\mathcal{A}^\dagger$. We then denote a linear differential expression $\mathcal{A}$ *formally self-adjoint* in case of $\mathcal{A}^\dagger = \mathcal{A}$. If $\mathcal{A}$ has constant coefficients, then $\mathcal{A}$ proves to be formally self-adjoint iff $c_\alpha \in \mathbb{R}$ for $|\alpha|$ even and $c_\alpha \in i\,\mathbb{R}$ for $|\alpha|$ odd.

In the following let us collect a couple of formally self-adjoint linear differential expressions and describe some self-adjoint realizations of them:

(a) *Sturm-Liouville operators.* Our first example of a formally self-adjoint linear differential operator (given in divergence form) is the *Sturm-Liouville operator*

$$\mathcal{A}_{sl}(u,x) \;:=\; -(p(x)\, u'(x))' \;+\; q(x)\, u(x), \qquad x \in (a,b)$$

with $p(x) > 0$ and $q(x) \geq 0$ for $x \in [a,b]$, $-\infty < a < b < \infty$. This operator determines the elongations of a vibrating string being fixed at the endpoints of $[a,b]$. Then the corresponding Dirichlet form reads as

$$B_{\mathcal{A}_{sl}}(u,v) \;=\; \int_a^b (p\,u'\,\overline{v}' \;+\; q\,u\,\overline{v})\,dx,$$

which shows that the expression $\mathcal{A}_{sl}$ is symmetric and half-bounded on the domain of definition $C_c^\infty(G)$. The energetic space $H_{\mathcal{A}_{sl}}$ is the Hilbert-Sobolev space $\overset{\circ}{H}{}^1_0(a,b)$ containing the restrictions of *absolutely continuous functions* $u:[a,b]\to\mathbb{R}$ with boundary conditions $u(a)=u(b)=0$ to the interval (a,b).

The self-adjoint extension $A_{sl,F}$ of the Sturm-Liouville operator in $L^2(a,b)$ has domain of definition

$$\mathfrak{D}(A_{sl,F}) = H^2(a,b) \cap \overset{\circ}{H}{}^1_0(a,b).$$

The operator $A_{sl,F}$ has pure point spectrum, since the interval (a,b) is assumed to be bounded. Especially in case of $p(x)=1$ and $q(x)=0$ for $x\in[a,b]$, the eigenvalues and eigenfunctions of the operator $\mathcal{A}_{sl,F} = \,-\,d^2/dx^2$ can exactly be determined. In fact, the eigenvalue distribution of $\mathcal{A}_{sl,F}$ is given by

$$\lambda_n \;=\; \frac{n^2\pi^2}{(b-a)^2} \qquad (n\in\mathbb{N}),$$

so the asymptotical behaviour of the eigenvalues is $\lambda_n \sim n^2$. Each eigenvalue λ_n owns a one-dimensional eigenspace, and the eigenfunction u_n of λ_n is given by

$$u_n(x) \;:=\; \sin\big(\frac{n\pi x}{b-a}\big) \quad \text{for } x\in[a,b].$$

Finally, the sequence of the eigenfunctions u_n constitutes after suitable renormations an orthonormal base in $L^2(a,b)$.

(b) *(Negative) Laplacian in various domains* $G\subset\mathbb{R}^d$. In order to apply the theory of self-adjoint operators to the investigation of the negative Laplacian, we start with the realization $A_{-\Delta,0}$ of the expression $-\Delta$ in the space $C_c^\infty(G)$ of test functions.

Using GREEN's second formula, and due to the boundary condition $\phi(x)=0$ for $x\in\partial G$ and $\phi\in C_c^\infty(G)$, we compute for $\phi,\psi\in C_c^\infty(G)$

$$\int_G -\Delta\phi\,\overline{\psi}\,dx \;=\; \sum_{i=1}^n \int_G \frac{\partial^2\phi}{\partial x_i^2}\,\overline{\psi}\,dx$$

$$=\; \sum_{i=1}^n \int_G u\,\overline{\frac{\partial^2\phi}{\partial x_i^2}}\,dx \;=\; \int_G \phi\,\overline{\Delta\psi}\,dx,$$

which implies symmetry of the operator $A_{-\Delta,0}$. Moreover, we obtain for $\phi \in C_c^\infty(G)$ the inequality

$$\langle -\Delta\phi, \phi\rangle = \sum_{i=1}^{n} \int_G \frac{\partial^2\phi}{\partial x_i^2}\,\bar{\phi}\,dx = \sum_{i=1}^{n} \int_G \left|\frac{\partial\phi}{\partial x_i}\right|^2 dx \geq 0,$$

hence $A_{-\Delta,0}$ is half-bounded with lower bound 0 and fulfills the prerequisites to apply the Friedrichs extension method. Thus we may quote that the operator $A_{-\Delta,0}$ owns at least one self-adjoint extension, and this assertion is true actually for arbitrary domains $G \subset \mathbb{R}^d$.

In the following enumeration we consider various self-adjoint realizations of the negative Laplacian in a domain G, all of them being extensions of the operator $A_{-\Delta,0}$:

(i) *(Negative) Laplacian in* $\mathbb{R}^d$. The analysis of the operator $-\Delta$ in the entire Euclidean space $\mathbb{R}^d$ is easy, since the differential operator proves to be unitarily equivalent to the multiplication operator A_m generated by the function

$$m : (x_1, \ldots, x_n) \mapsto (x_1^2 + \ldots + x_n^2) = |x|^2 \quad (x \in \mathbb{R}^d).$$

In fact, the operator A_m is real-valued and self-adjoint, which is a consequence of the *Fourier-Plancherel transformation*[4]. This implies that the operator A_{sa} with domain of definition $\mathfrak{D}(A_{sa}) := H^2(\mathbb{R}^d)$ must be self-adjoint, too. The spectrum of A_{sa} is that of the operator A_m, that is, it is purely continuous and $\sigma_c(A_{sa}) = [0, \infty)$. Moreover, since A_{sa} is the closure of the operator $A_{-\Delta,0}$, the latter proves to be essentially self-adjoint.

(ii) *Dirichlet-Laplacian.* Let the domain $G \subset \mathbb{R}^d$ be *bounded* and

[4] We introduce the *Fourier-Plancherel transformation* in $L^2(\mathbb{R}^d)$ by

$$\hat{u}(\xi) := \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} e^{-2\pi i\langle \xi, x\rangle}\, u(x)\, dx$$

for all $u \in C_c^\infty(\mathbb{R}^d)$, which forms a partial isometry $\mathscr{F} : u \mapsto \hat{u}$ from the space $C_c^\infty(\mathbb{R}^d)$ to the space $L^2(\mathbb{R}^d)$. Then for functions $u, v \in L^2(\mathbb{R}^d)$ the inner product between u and v and the inner product of their Fourier-Plancherel transforms $\hat{u}$ and $\hat{v}$ are related by PLANCHEREL*'s formula* $\langle u, v\rangle = \langle \hat{u}, \hat{v}\rangle$. Since the mapping $\mathscr{F}$ is linear and continuous on $C_c^\infty(\mathbb{R}^d)$, which is a dense subspace of $L^2(\mathbb{R}^d)$, it can be extended to an *unitary*, that is, a bijective and isometric linear operator $\mathscr{F}_P : L^2(\mathbb{R}^d) \to L^2(\mathbb{R}^d)$, called the *Fourier-Plancherel operator* or the L^2*-Fourier transformation* in $L^2(\mathbb{R}^d)$.

consider $-\Delta$ with domain of definition $\mathfrak{D}(-\Delta) := C_c^\infty(G)$. The related Dirichlet form $B_{-\Delta}$ is given by

$$B_{-\Delta}(\phi,\psi) := \int_G \nabla\phi\,\nabla\psi\,dx \qquad (\phi,\psi\in C_c^\infty(G)),$$

which can continuously be extended to the sesquilinear form $\overline{B_{-\Delta}}$ with domain of definition $\mathfrak{D}(\overline{B_{-\Delta}}) := H_0^1(G)\times H_0^1(G)$.

Due to theorem 10.5, the operator A_D associated with the form $B_{-\Delta}$ and defined by

$$\begin{cases} \mathfrak{D}(A_D) := \{u\in H_0^1(G) : \Delta u\in L^2(G)\} \\ A_D u := -\Delta u \text{ for } u\in\mathfrak{D}(A_D) \end{cases}$$

is self-adjoint, called the *negative Laplacian with Dirichlet boundary conditions* or briefly the *Dirichlet-Laplacian.*

We obviously have $H_0^1(G)\cap H^2(G)\subset\mathfrak{D}(A_D)$, and one can prove that if G is bounded and owns a C^2-boundary or $G=\mathbb{R}_+^n$, then the reversed inclusion also holds, hence $\mathfrak{D}(A_D)=H_0^1(G)\cap H^2(G)$. This equality is even true in the case, where G is bounded and convex.

Furthermore, if G has the 1-extension property, then the space $H_0^1(G)$ is compactly embedded into the space $L^2(G)$ and A_D is an operator with pure point spectrum. In this case it is an important property of the Dirichlet-Laplacian that the number 0 does *not* belong to the spectrum $\sigma(A_D)$, so A_D is invertible and the operator A_D^{-1} proves to be bounded.

(iii) *Neumann-Laplacian.* Let $G\subset\mathbb{R}^d$ be a bounded domain with C^1-boundary ∂G and let the realization A_N of the negative Laplacian in $L^2(G)$ be defined by

$$\begin{cases} \mathfrak{D}(A_N) := \{u\in H^2(G) : u_v(x)=0,\ x\in\partial G\} \\ A_N u := -\Delta u \text{ for } u\in\mathfrak{D}(A_N). \end{cases}$$

The realization A_N is said to be the *(negative) Laplacian with Neumann boundary condition* or briefly the *Neumann-Laplacian.* It owns $\lambda=0$ as the minimal eigenvalue, and all constant functions in $\mathfrak{D}(A_N)$ are eigenfunctions of $\lambda=0$.

(c) *Schrödinger operators.* By the *Schrödinger operator* is meant the linear partial differential expression given by

$$\mathcal{H}_q(u,x) := (-\Delta u)(x) + q(x),$$

defined in the entire Euclidean space $\mathbb{R}^d$. In the special case, where $q=0$, self-adjointness of the operator $-\Delta$ acting in $L^2(\mathbb{R}^d)$ and with domain of definition $H^2(\mathbb{R}^d)$ is well-known.

The so-called *potential function* $q:\mathbb{R}^d \to \mathbb{R}$ gives rise to a multiplication operator M_q acting in $L^2(\mathbb{R}^d)$, that is,

$$\begin{cases} \mathfrak{D}(M_q) := \{u \in L^2(\mathbb{R}^d) : q\cdot u \in L^2(\mathbb{R}^d)\}, \\ M_q u := q\cdot u \ \text{ for } u \in \mathfrak{D}(M_q), \end{cases}$$

so the operator $\mathcal{H}_q$ can be seen as a perturbation of the negative Laplacian by the operator M_q.

By the time-dependent *Schrödinger equation* is meant the initial value problem

$$i\,\dot{u} = -\Delta u + M_q u$$

with prescribed initial value $u(0)=u_0 \in H^2(\mathbb{R}^d)$, where it is supposed that $\mathcal{H}_q = -\Delta + M_q$ is a self-adjoint operator in $L^2(\mathbb{R}^d)$ with domain of definition $\mathfrak{D}(\mathcal{H}_q)=H^2(\mathbb{R}^d)$. The existence and uniqueness of a solution for this problem is ensured by the functional calculus for self-adjoint operators using the spectral resolution of the operator $\mathcal{H}_q$ and the function $f : \lambda \in \mathbb{R} \mapsto e^{it\lambda}$. The solution u of the problem is then provided by a group of unitary operators $\{U(t)\}_{t\in\mathbb{R}}$, which implies $u(t)=U(t)u_0$ for every $t\in\mathbb{R}$ (see also section 5 for the treatment of the abstract Schrödinger equation).

The study of *time-independent solutions* of the Schrödinger equation leads to the eigenvalue problem

$$\mathcal{H}_q u := -\Delta u + M_q u = \lambda u, \tag{12.12}$$

which means to find those real numbers λ, for which there are non-trivial solutions of (12.12). The discrete spectrum $\sigma_d(\mathcal{H}_q)$ of the operator $\mathcal{H}_q$ is in fact of great interest, since the eigenfunctions of eigenvalues $\lambda \in \sigma_d(\mathcal{H}_q)$ describe the stationary states of a quantum mechanical system.

The presence of a potential function q can fairly change the spectrum of a Schrödinger operator like the example of the *harmonic oscillator* shows: the spectrum of the negative Laplacian is the connected set $[0,\infty)$, but the spectrum of the operator $-\Delta + M_q$ with $q : x \mapsto \frac{1}{2}x^2$ consists of discrete points only.

It is a further and interesting problem in quantum mechanics to determine lower and upper bounds for the discrete spectrum of the operator $\mathcal{H}_q$. In the following we want to present three assertions, which contain information on the bounds of the set $\sigma_d(\mathcal{H}_q)$, namely the *Birman-Schwinger bound*, the *Lieb-Cwikel-Rozenblum bound* and KATO*'s theorem*:

1. The *Birman-Schwinger bound.* This assertion presents an upper bound for the set $\sigma_d(\mathcal{H}_q)$, which is valid for certain potentials q in case of $d=3$. Let $q \in L^\infty(\mathbb{R}^3)$ and
$$B_q := \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \frac{q(x)\, q(y)}{|x-y|^2}\, dx\, dy < +\infty.$$
Then the total number $N(\mathcal{H}_q)$ of eigenvalues of the operator $\mathcal{H}_q$ can be estimated according to
$$N(\mathcal{H}_q) \leq \frac{1}{16\pi^2} B_q.$$
2. The *Lieb-Cwikel-Rozenblum theorem* quotes an estimate on the number $N(\mathcal{H}_q)$ of negative eigenvalues of the operator $\mathcal{H}_q$: if $d > 3$ and $q \in L^\infty_{loc}(\mathbb{R}^d)$, then
$$N(\mathcal{H}_q) \leq c_d \int_{\mathbb{R}^d} \min\{q(x), 0\}^{n/2}\, dx,$$
where c_d is a constant only depending on the dimension d.
3. Our last result is often denoted as *Kato's theorem.* It does not make assertions on bounds for the discrete spectrum, but quotes the absence of positive eigenvalues under mild assumptions on the function q: if $q \in L^\infty_{loc}(\mathbb{R}^d)$ and
$$\lim_{|x|\to\infty} |x|\, q(x) = 0,$$
then the operator $\mathcal{H}_q$ does not own any positive eigenvalue.

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